Causal Nowcasting of 10 ms Charge-Exchange Spectroscopy Gaps in KSTAR: an LSTM with Lower Error than Acausal Interpolation and the 20-Parameter Physical Filter It Computes

Seungsang Lee
Department of Nuclear Engineering, Seoul National University
lss010330@snu.ac.kr

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Abstract

Charge-exchange spectroscopy (CES) is the primary measurement of ion temperature (T_i) and toroidal rotation ($V_{\rm rot}$) in tokamaks, but photon collection makes it slow and intermittently missing: on the KSTAR 10 ms grid studied here (641 discharges), 8.2% of T_i and 23.9% of $V_{\rm rot}$ samples are absent, the two targets go missing independently, and a further 41.1% of $V_{\rm rot}$ rows are instrument holds, leaving 65.0% of the grid without independent $V_{\rm rot}$ information. The incumbent way to fill a gap is interpolation of the CES channel itself, which reads neighbours on both sides of the gap and is therefore unavailable online. We ask whether a strictly *causal* estimator—the always-available fast diagnostics (beam-emission spectroscopy, electron-cyclotron-emission imaging, Mirnov coils) plus the irregular past CES history—can fill a missing 10 ms row with lower error than that future-using interpolation. The estimator is a compact two-branch causal LSTM (0.36M parameters) that reads every grid row of a discharge segment, supervises the sparse targets through a per-target masked loss, and routes the $V_{\rm rot}$ branch away from the fast diagnostics by construction. Under a pre-registered protocol—hold-free training and scoring, file-level splits, a test set never read by model selection, a shot-clustered paired bootstrap (10,000 resamples), and two co-primary populations that treat the 0.53% of T_i spectral-fit failures (>3 keV) as missing or keep them—the nowcaster's skill over monotone-cubic PCHIP interpolation on T_i is +0.17 to +0.26 (cut) and +0.23 to +0.32 (inclusive) with the 95% interval excluding zero on all four held-out splits in both populations, +0.08 to +0.17 over a causal Gaussian process with the interval excluding zero on 8/8 cells, and −0.05 to +0.11 against the offline GP with the interval including zero on 7/8; a pre-registered shot-level repeated 5-fold replicate (100 runs, every discharge tested once) puts the overlap-corrected interval at $[+0.19, +0.25]$ (cut) and $[+0.19, +0.31]$ (inclusive). The skill survives two stress tests: reweighting onto the covariate distribution of the genuinely missing rows keeps the skill over persistence at +0.28 to +0.44 with intervals above zero on 8/8 cells, and a strict time split across campaigns keeps +0.17 to +0.20 over PCHIP with intervals above zero on 4/4 initialisations in both populations. The converse holds for rotation: zeroing the fast diagnostics leaves the $V_{\rm rot}$ output bit-identical, its mean-squared-error skill over PCHIP is tail-carried ($[+0.07, +0.70]$ cut under the registered two-level bootstrap), and in the median its error ratio to persistence is 0.94–1.00 with the interval including 1 on 3/4 splits—an information asymmetry we attribute to the unobserved NBI torque and to the 10 ms photon floor, stated for standard CES at 10 ms from the core to the pedestal top. We then ask what the nowcaster computes. A pre-registered four-stage readout (frozen-row equation, trained equation, full-row Jacobian, corrected form) writes the T_i branch as a 20-parameter physical filter: an observation filter whose gain rises with the gap since the last measurement ($K_0 = 0.27$–0.30 fresh, $K_\infty = 0.48$–0.53 after a long gap, transition $s_K = 39$–47 ms) plus a linear response to each fast channel's deviation from its preceding 100 ms baseline that is switched off beyond $\theta = 2.9$–3.0σ, with $\sum a_{\rm BES} = +48$ to $+53$ and $\sum a_{\rm ECEI} = +44$ to $+50$ eV per σ. Trained from scratch under the identical protocol, that equation keeps 72–78% of the backbone's PCHIP-relative skill and 87–90% of its persistence-relative skill on 4/4 cut splits (registered bar 0.70) and sits above the earlier increment-form equation by a paired +0.034 to +0.042 with intervals above zero on 4/4; the remaining 22–28% is the network's nonlinear event handling, which no low-order function of the

fast-channel history reproduces. Two secondary results separate estimands: a pre-registered hold-or-jump filter with Student-t noise at the training bulk scale is the median estimator ($0.91\times$ persistence for T_i on 8/8 split–population cells) while the nowcaster is the mean estimator, and a Student-t distribution head is calibrated (CRPS gain +0.229 for T_i and +0.363 for V_{rot} over the point reference, 50 %/90 % coverage 0.49–0.51 / 0.91–0.92, cut). The aggregate skill is a tail statistic, the offline claim is bounded above by the offline GP, the value cut is a proxy for fit quality, and everything is one device and one campaign. We release the pipeline, the pre-registrations and the evaluation artifacts.

1 Introduction

Ion temperature T_i and toroidal rotation V_{rot} at the plasma edge govern key elements of tokamak performance and stability, including pedestal structure, ELM behavior, and access to high-confinement regimes. The workhorse measurement of both quantities is charge-exchange spectroscopy (CES) [1], which infers T_i from Doppler broadening and V_{rot} from the Doppler shift of charge-exchange line emission. Because CES must integrate photons to reach adequate signal-to-noise, it is intrinsically slow, and individual time points are frequently lost to exposure and signal-quality problems. In the KSTAR data studied here, even on a uniform 10 ms grid, 8.2% of T_i and 23.9% of V_{rot} observations are missing—*independently* of each other—while a further 41.1% of V_{rot} rows repeat the previous reading bit-for-bit (§3.4), so only 35.0% of the grid carries an independent V_{rot} measurement. Empirically in our dataset, the missingness concentrates at physically interesting moments (low signal-to-noise phases, ELMs, transitions), i.e. it is missing-not-at-random in the sense of Little and Rubin [2] (MNAR).

Meanwhile, several fast diagnostics observe the same plasma without gaps: BES measures density-fluctuation structure, ECEI images electron temperature, and Mirnov coils record magnetic fluctuations. None of them measures T_i or V_{rot} directly. The question this paper asks is:

> *At a 10 ms timestep where CES is missing, can the simultaneous fast diagnostics plus the irregular past CES history recover T_i and V_{rot} information that temporal interpolation of CES alone cannot?*

A positive answer would justify a data-driven *virtual sensor*: a model that fills CES gaps online, without the strong inverse-mapping assumptions required by hardware alternatives, enabling gap-free pedestal analysis and, ultimately, real-time use.

A deliberately hard bar. The obvious baseline for gap-filling is interpolation of the CES channel itself. We make the comparison intentionally adverse to the model: the interpolation baselines (linear, monotone-cubic PCHIP [3], a local AR reference, and a Gaussian process) are *offline*—they may use CES neighbours on *both sides* of the target time, including the future—whereas the model is strictly causal, seeing only the fast diagnostics up to the target step and the past CES history. If a causal model beats a future-using interpolant, the fast diagnostics must carry CES-relevant information that the temporal structure of CES alone does not contain; and a model that beats future-using interpolation a fortiori beats every causal baseline. Because the strongest *deployable* competitor is not persistence but a past-only smoother, we also add a *causal* GP arm and report the model against it separately.

Contributions. The argument runs in one direction: the problem (missing CES rows, a causal nowcast against acausal interpolation), the T_i result and its replication, the $T_i \leftrightarrow V_{\mathrm{rot}}$ asymmetry as the converse evidence, the equation the nowcaster computes, and the two estimands it does not fold into one output.

1. **A causal nowcast of missing 10 ms CES ion temperature with lower error than acausal interpolation, replicated** (§6.2). The skill over future-using PCHIP is +0.17 to +0.26 (cut) on four held-out splits, and its overlap-corrected 95% interval in a pre-registered repeated 5-fold replicate is $[+0.19, +0.25]$, excluding zero.
2. **The equation the nowcaster computes, registered and retained** (§4.4). A four-stage pre-registered readout writes the T_i branch as a 20-parameter physical filter which, trained from scratch under the identical protocol, keeps 72–78% of the backbone's PCHIP-relative skill on 4/4 cut splits (registered bar 0.70).

3. **The $T_i \leftrightarrow V_{\mathrm{rot}}$ information asymmetry as the converse evidence** (§6.7, §6.11). Zeroing the fast diagnostics leaves the V_{rot} output bit-identical; its mean-squared-error skill over PCHIP has the registered two-level 95% interval $[+0.07, +0.70]$ (cut), excluding zero and tail-carried, and the statement is scoped to standard CES at 10 ms with the torque unobserved.

4. **Mean and median from separate estimators, and a calibrated distribution** (§6.10, §6.11, §6.12). A pre-registered hold-or-jump filter with Student-t noise has a T_i median-error skill over the backbone of $+0.06$ to $+0.39$, with the 95% interval excluding zero on $4/4$ splits in both populations; the nowcaster is the mean estimator and a Student-t head carries the calibrated uncertainty.

5. **Two stress tests that separate the offline from the deployment claim** (§6.5, §6.6). Reweighted onto the genuinely missing rows, the T_i skill over persistence is $+0.28$ to $+0.44$ with the 95% interval excluding zero on $8/8$ split–population cells; the skill over PCHIP also keeps its interval above zero across a strict campaign split.

6. **The ceiling is the input set, not the estimator** (§6.9, §6.8, §6.14). A 21,498-parameter persistence-plus-bounded-correction model has a paired T_i skill against the backbone of $+0.002$ under the cut, every split's 95% interval including zero; width, model family and context beyond 50 ms are closed axes.

7. **One evaluation methodology, described in full in §5**: pre-registration with a scored prediction ledger per batch, two co-primary populations, a shot-clustered paired bootstrap, the Nadeau–Bengio overlap-corrected interval and a two-level (training repeat × discharge) bootstrap. Each piece has a precedent in another field (§2); it is reported as a combination, not as new.

2 Related work

Profile fitting and gap-filling in fusion diagnostics. Gaussian-process regression [4] is the community-standard tool for fitting and uncertainty-quantifying fusion profile data [5, 6] — applied to KSTAR's own CES T_i profiles with SVM-based outlier rejection [7] — and monotone interpolants such as PCHIP [3, 8] are preferred over natural cubic splines [9] near transients because splines exhibit Gibbs-like ringing at ELM- and sawtooth-like steps [10]. Irregularly sampled series admit exact AR-type null models (IAR/CIAR) [11, 12]. We use this ladder—persistence, linear interpolation [13], PCHIP, local AR—as pre-registered baselines, with PCHIP as the headline comparator, and Murphy's MSE skill score [14] as the metric. A persistence baseline has precedent in fusion profile forecasting—Abbate et al. [15] report that both machine-learning and physics-based predictors, extrapolated out of distribution, fall below the assumption that the profile does not change—and skill scores against a persistence reference are standard in space-weather forecasting [16], but we found no fusion precedent for Murphy's skill against persistence and against acausal interpolation as the evaluation bar, nor for the term *nowcasting* in its meteorological sense. Our contribution is orthogonal to this literature: rather than a better CES-only interpolant, we test whether *other diagnostics* add information beyond any CES-only method, by benchmarking a causal multimodal model against future-using interpolation.

Neural diagnostic inference in fusion. Neural diagnostic inference has three main lines, and it is instructive to ask of each: *what happens when the target measurement is absent?* The oldest line accelerates spectral fitting—NN analysis of JET CX spectra [17, 18] and fast T_i estimation from EAST CXRS spectra [19], and the newest member, on *our* diagnostic: NN-CES, a physics-constrained network that extracts T_i and rotation from KSTAR CES spectra without beam modulation [20]. These gain orders of magnitude in speed over least-squares fits, but by construction they require measured spectra at the timestep: a missing CES exposure leaves them with no input at all. NN-CES is the closest prior work, so the distinction fits in one sentence: *NN-CES reads the measured timesteps better; this work fills the ones that are not measured.* A second line replaces one inference pipeline with another diagnostic's: Bayesian-model emulation at W7-X [21], profile shapes from scalar parameters on NSTX-U [22], and—closest to our targets—T_i and rotation profiles from an x-ray crystal spectrometer on EAST [23]. Lin et al. predict exactly our two quantities, but from another Doppler spectrometer, i.e. the same physical information channel with the same photon-statistics limits; the mapping is instantaneous and inherits the source diagnostic's availability. A third line builds end-to-end surrogates for slow reconstruction codes: bolometer tomography [24], line-integral diagnostics [25],

safety-factor profiles [26], real-time equilibria [27], and—on KSTAR itself—NN Grad–Shafranov solvers [28, 29] and rapid EPED-parameterized profile reconstruction [30] — the latter consuming CES as an *input* and explicitly naming rotation as future work. These address analysis *latency*, not measurement *availability*: they accelerate inversions of data that exist. Two works sit closer to the availability question. Actuator-conditioned profile forecasting on DIII-D [31] is causal and history-driven like our model, but forecasts from *dense* measured profiles for control, rather than recovering a sparse channel. RTCAKENN [32] reconstructs kinetic profiles (including impurity temperature and rotation) in real time and keeps operating when Thomson/CER inputs drop out, via dropout training; missingness there is a robustness condition handled by zeroing inputs, whereas in our setting the target's own sparsity is the problem itself, tracked explicitly through per-target observation flags and irregular-time features. Across all three lines the common assumption is that the information source is available when the model runs; recovering a diagnostic at timesteps where *its own* measurement is missing, from other channels plus its past, falls outside their scope.

Cross-diagnostic reconstruction and temporal densification (closest prior work). A recent family reconstructs one diagnostic from others. GP-based imputation restores faulty magnetic sensors on KSTAR in sub-ms [33]; ML detects and compensates corrupted sensors [34]; autoencoders fuse redundant density diagnostics for resilience [35]; time-series extrinsic regression fills missing electron-temperature data on EAST [36], and a CNN–BiGRU–transformer reconstructs missing effective charge Z_{eff} on EAST from exogenous signals [37]—bidirectional, hence acausal, and described by its authors as offline signal restoration; and FusionMAE performs masked-channel "virtual backup diagnosis" across 88 channels on HL-3 [38]. Closest to our framing, Diag2Diag synthesizes Thomson-scattering T_e/n_e at fluctuation-diagnostic rates from other simultaneous diagnostics on DIII-D [39], and temporal super-resolution of Thomson profiles from fast radiation diagnostics has been demonstrated on COMPASS [40]; a model-based observer on TCV fills the intervals of low-rate Thomson scattering with high-rate interferometry in an extended Kalman filter, causally and for electron density [41]; PanoMHD models multimodal signal dynamics with a causal transformer for control [42]. Our work differs from this family on four axes, each verifiable from the cited works' stated scope: (i) *target channel*—every member reconstructs electron quantities or generic dense channels (Thomson T_e/n_e in 39 and 40; ECE-type T_e in 36; arbitrary masked channels in 38), and where CER/CES appears at all it is an *input* [39], never the sparse target; (ii) *causality*—Diag2Diag (a memory-less multilayer perceptron that maps same-instant inputs to each Thomson time point and is scored against the measured points with no interpolation or persistence baseline) and the COMPASS study are simultaneous mappings, and FusionMAE reconstructs within a window, whereas our model conditions on the target's own irregular past and never reads the future; (iii) *evaluation bar*—none benchmarks against *future-using* interpolation of the target channel, which for gap-filling is the incumbent method to beat, under a pre-registered shot-clustered protocol; and (iv) *information analysis*—reconstructability is assumed rather than tested per target, whereas the $T_i \leftrightarrow V_{\text{rot}}$ asymmetry is our central finding. In sum, this work continues the programme this active family has established—reconstructing an absent measurement from the diagnostics that are always on—and extends it in three directions: from electron channels to a *sparse ion channel*, from simultaneous memory-less mappings to *causal estimation conditioned on the target's own irregular past*, and from assumed reconstructability to *per-target tests against pre-registered baselines*. A literature check (OpenAlex, arXiv, Crossref; 2026-11-09; full-text search incomplete) found no precedent for the conjunction of these three—causal temporal gap-filling of sparse CXRS ion measurements—on DIII-D, JET, ASDEX Upgrade, EAST or KSTAR; we present it as the family's natural next step rather than as a departure from it, and "not found" is not "does not exist".

Irregular-time and masked multitask learning. Our design borrows standard ingredients from the irregular time-series literature—observation masks and decay-aware recurrence (GRU-D [43]), attention over continuous-time embeddings (mTAN [44]), and latent-ODE dynamics [45]—together with multitask learning under partial labels [46], late-fusion multimodal architectures [47, 48], pre-LayerNorm transformers [49], and attention pooling [50]. The "nowcasting" framing follows its use in machine-learning weather prediction [51, 52]. The novelty is not any single ingredient but the leakage-hardened combination under a physics-adverse evaluation protocol.

Methods lineage. Each method we lean on is standard in another field, and we cite its source rather than claim it. Fixing the analysis plan before the data are read follows the registered-report logic of Nosek and

Lakens [53] and its adoption in machine learning by the NeurIPS pre-registration workshops [54]. The overlap correction for repeated cross-validation is Nadeau and Bengio [55], and the reason a plain cross-validation interval under-covers is given by Bates et al. [56]. Resampling discharges as clusters, with the training repeat as an outer level, is the hierarchical bootstrap of Saravanan et al. [57] and Ren et al. [58], with the structure of the Multi-Bootstrap over seeds and test examples [59] and of the start-date and ensemble-member bootstrap for skill scores in decadal prediction [60]; what we did not find is this combination (cross-validation repeats × discharge clusters × skill ratio). The skill score is Murphy's [14]; proper scoring of the predictive distribution follows Gneiting and Raftery [61], and a likelihood-fitted Student-t predictive distribution scored by the CRPS, with a network that outputs its parameters, comes from weather-ensemble post-processing [62, 63]. Reading a trained recurrent network through its input–output response rather than its weights follows Sussillo and Barak [64], and turning a trained network into a closed-form equation that is then tested on its own follows Cranmer et al. [65]; the retention test is the procedure of Maheswaranathan and Sussillo [66] ("recovering greater than 90% of the RNN's performance increase over the baseline") in the non-inferiority form of retaining a prespecified fraction of an effect [67], and what we did not find is the two combined with a threshold committed in advance, or either on a fusion predictor. That weight magnitude is not importance is the result of optimal brain damage [68], and Allen and Bishop [69] reached the same conclusion for a tokamak confinement network in 1992, reading its response instead of its weights. The Student-t measurement model of the physics filter is the robust Kalman filtering of Roth et al. [70] and Agamennoni et al. [71]. Fusion diagnostic time series have precedents for a state-space Student-t process solved by Bayesian filtering (DIII-D; 72), an interacting-multiple-model estimator (ASDEX Upgrade ECE; 73) and an ensemble Kalman filter on measured ion and electron temperature (LHD; 74); what we did not find is a hold-or-jump prior with Student-t noise on charge-exchange time series. In peer-reviewed fusion machine learning the check found test discharges resampled identically across models, reported as medians and interquartile ranges [75], but no interval on a paired skill difference clustered by discharge, no pre-registered analysis plan, no Murphy-form skill against persistence or interpolation and no response-based readout of a recurrent network—"no precedent found", not absence: the check (OpenAlex with full-text search, arXiv, Crossref; 2026-11-10) read mostly abstracts.

3 Data and problem setup

3.1 Dataset

We use 641 KSTAR discharge (shot) files, shots 30801–32751, selected by the data provider for hardware consistency and for H-mode ELM-suppression (RMP) phases, and aligned to a common 10 ms grid: 247,207 rows in total (median 339 per file). Each row provides BES [76] (9 channels), ECEI [77] (4 channels), Mirnov coils (2 channels), time, and the two CES targets T_i and V_{rot} [1]. On this grid T_i is missing in 8.2% of rows and V_{rot} in 23.9%, independently (Fig. 1). Rows are grouped into contiguous segments at time gaps ≥ 0.5 s (a threshold that sits in the valley of a bimodal delta distribution: only 82 of the ∼247k inter-row deltas fall in $(0.1, 0.5)$ s; within a segment 99.4% of steps are 10 ms): a typical file holds **one main segment** (median 301 rows ≈ 3.0 s; 10th–90th percentile 1.3–7.0 s; 28 files have two and 20 have none longer than 10 rows) plus a few isolated single rows (1,279 across the set, typically the first row at $t = 0$) separated from it by gaps of median 6.3 s. No arm—neither an interpolant nor a model's input—bridges a gap.

3.2 Two framings of a training example

The *windowed* framing (our control, §4) builds a sample at target index t from the W preceding grid rows: `bes` $(W, 9)$, `ecei` $(W, 4)$, `mc` $(W, 2)$, `time_features` $(W, 4)$, `ces_history` $(W, 4)$, the normalized target $[T_i, V_{\mathrm{rot}}] \in \mathbb{R}^2$, and a per-target observation mask $m \in \{0, 1\}^2$. The four time features—lookback seconds, inter-row delta seconds, and their $\log(1+x)$ transforms—expose the irregular observation pattern explicitly (the reliability of a past CES value depends strongly on whether it is 10 ms or 200 ms old); the four CES-history channels are the previous normalized T_i, previous normalized V_{rot}, and one observation flag per target. W=2 throughout (§6.14).

The *full-grid sequence* framing (our main model) keeps every input-complete row of a contiguous segment—labelled or not—as context, because the fast diagnostics have no missing values. Each step carries 22 strictly

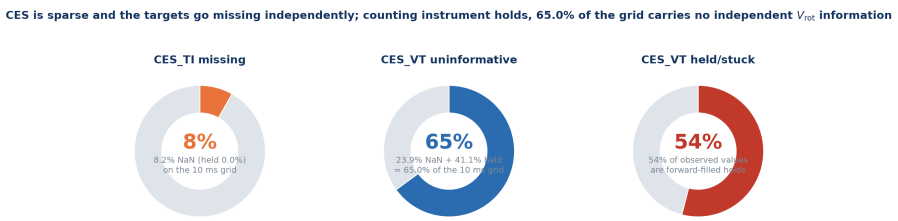


Figure 1: CES missingness structure on the 10 ms grid. T_i and V_{rot} go missing independently (8.2% / 23.9%), motivating per-target masking rather than row filtering; counting instrument holds, 65.0% of the grid carries no independent V_{rot} information.

causal channels: the 15 z-scored fast channels, $\log(1+\Delta t_{\mathrm{prev}})$, and per target the last observed value carried forward from rows $< t$, log(1+staleness), and a has-any-past-observation flag. Targets are z-scored with a parallel observed mask, and the sparsity is handled entirely in the loss. Nothing in either framing lets a model read the target row's own value.

3.3 Design principles

No fabricated labels. We never impute targets to create training rows. In the windowed framing a row is kept iff its diagnostic inputs are complete and *at least one* CES target is genuinely observed; in the sequence framing unlabelled rows contribute context only.

Per-target masked loss. The loss is masked per target, $\mathcal{L} = \sum_{k\in\{T_i,V_{\mathrm{rot}}\}} m_k\,(\hat{y}_k - y_k)^2 / \sum_k m_k$, so a row with only one observed target still supervises that target. A both-targets-required filter would silently discard ≈28% of labelled rows; removing it is a pure data gain.

Leakage hardening. (i) *File-level splits*: adjacent rows within a shot are strongly autocorrelated, so splits are made at the shot-file level and pinned to disk; (ii) *train-file-only normalization*: all z-score statistics (NaN-aware for the sparse targets) are estimated on training files only—the sequence model additionally standardizes each discharge's fast diagnostics by that discharge's own mean and variance, which uses no other discharge and no target (§6.6 shows why this matters); (iii) *target-step masking*: the target row's own values and observation flags never enter the input.

3.4 Data-quality audit 1: held (forward-filled) V_{rot} values

54% of observed V_{rot} values are instrument holds: bit-identical repeats of the previous observation within a contiguous time block (runs up to 1,214 rows; 499/641 files affected). These arise because the native V_{rot} cadence is slower than the row cadence; they are not independent measurements. V_{rot} is recorded to five decimal places with a minimum spacing of 4×10^{-5} between distinct values, so "consecutive-equal" has no measurable false-positive channel; T_i is unaffected (1 held row in 226,991). Under the confirmed protocol holds are removed *everywhere*—from the supervision targets, from the history/carried inputs and their observation flags, from the normalization statistics, and from the interpolation anchors of every baseline—so no arm can score a forward-fill for free. Every number in this paper is on genuine measurements only, and hold-free is the protocol rather than a sensitivity row because holds also contaminate *training* (a forward-filled staircase teaches a model that copying history is near-optimal).

3.5 Data-quality audit 2: T_i spectral-fit failures and the two populations

Observed T_i has p99 = 2,089 eV, p99.9 = 9,601 eV and a maximum of 14,984 eV: the far tail is not plasma physics but failed spectral fits. **1,197 rows (0.53%) exceed 3 keV**, forming 951 runs in 274 discharges; 85% of the runs are single rows, only 2% last five rows or more, and the median run peak is 13× the mean of its observed neighbours (IQR 6–26×). Such rows are unpredictable for every method, and—worse for a fair comparison—a spike *used as an interpolation anchor* poisons the interpolant at its neighbours. Two

responses are defensible and each is open to a criticism: treating >3 keV as missing in every arm removes rows no method can predict but invites "the hard rows were removed", while keeping them handicaps the offline baseline by feeding it spiked anchors. We therefore pre-registered **two co-primary populations**: *cut* (values >3 keV treated as missing at load time, identically in supervision, history inputs, normalization statistics and every baseline's anchors) and *inclusive* (no cut). **A claim is stated without qualification only if it holds in both**; a result that holds in one is reported with its population named. The form is that of non-inferiority trials, where the full analysis set and the per-protocol set carry equal importance and are expected to lead to similar conclusions [78]; the two populations define different estimands [79], so neither is read as a sensitivity analysis of the other. The threshold is immaterial across 2.5–4 keV (§6.13); the value cut is a physically motivated proxy (">3 keV is not a KSTAR ion temperature") pending CES fit-quality metadata, and it is one-sided—the same audit finds single-sample *downward* dips (4,965 rows $\leq \frac{1}{2}$ both neighbours) that no value cut touches, and the cut removes only 19% of the $\geq 2\times$ upward outliers. V_{rot} has fit-failure spikes too (119 rows above 1,000 km/s in 16 shots, 101 of them one block in a single discharge); the protocol leaves V_{rot} uncut and every persistence-anchored V_{rot} comparison reports the share of squared error those rows carry.

4 Model

The main model, `seq_v2`, is a **357,570-parameter** causal sequence nowcaster; the windowed model it replaces is a 201,258-parameter late-fusion network and is retained as the paired control. Before describing either, this section fixes three things: *what is being estimated*, *why those inputs are wired to that target*, and *which time scales the model sits on*. None of the three is measured by the results section; they are what makes those measurements interpretable, and for the routing they make it a consequence of a conservation law rather than a modelling preference.

4.1 The estimand, the loss, and an irreducible floor

At each grid time t the model estimates the normalized target $y_t = [T_i, V_{\mathrm{rot}}]_t$ from the observation history $x_{\leq t}$. The loss is a masked mean squared error over per-target observation masks $m_{t,c} \in \{0, 1\}$, plus a physical penalty discouraging T_i predictions below the normalized zero z_0:

$$\mathcal{L} = \frac{\sum_t \sum_{c \in \{T_i, V_{\mathrm{rot}}\}} m_{t,c} \left(\hat{y}_{t,c} - y_{t,c}\right)^2}{\sum_t \sum_c m_{t,c}} \; + \; \lambda \, \overline{\mathrm{ReLU}\left(z_0 - \hat{y}_{t,T_i}\right)}, \qquad \lambda = 0.1. \tag{1}$$

The minimizer of the first term of (1) is the conditional expectation $\mathbb{E}[y_t \mid x_{\leq t}]$, so what this work reports is the mean of the conditional distribution and not the distribution itself. Uncertainty is handled separately, by a predictive-distribution head (§6.12).

This objective has a floor no estimator can remove. The target is the output of a photon-integrated spectral fit, so it carries its own measurement noise σ_{meas}, and even a predictor that recovers the true physical quantity exactly would score

$$\mathrm{RMSE}^2 \; = \; \sigma_{\mathrm{meas}}^2 \; + \; b^2 \tag{2}$$

against it, where b is the physical error. Difference-based estimators applied to the observed series put σ_{meas} for T_i in the cut population at **46.4–129.9 eV** (fourth-difference bound 129.9, trimmed at the 95th percentile 68.7, robust MAD 46.4). The backbone's RMSE of 157.8 eV is the same order as that range, and the reason its two ends differ by nearly a factor of three is what decides the character of the loss: **46.6% of the fourth-difference squared mass (65.6% for V_{rot}) sits in the top 1% of one-step changes.** The aggregate in (1) is therefore a *tail statistic*, and the later finding that width and operator family are flat (§6.8) should be read as a consequence of that fact rather than as an exhaustion of information: in the bulk the model is still 2.3–3.4× above the target's own scatter.

$\lambda = 0.1$ is inherited from the windowed pipeline rather than derived here, and the way to turn it into a derivation is to fix a target violation rate and solve for the multiplier as a constrained optimization. This version instead reports only that the penalty does not manufacture the result: every arm on the ladder is trained with the same λ, and the interpretable rung `b3k8`, which starts at persistence by construction and therefore almost never activates the penalty, matches the backbone (§6.8).

Table 1: Term-by-term audit of (4). Full 641-shot census, confirmed data treatment (spikes cut before held removal), 624 contiguous blocks. r is the lag-0 pooled Pearson correlation between a modality's standardized channel mean and the target, or its one-step increment.

Term	Physics	In our data?	Disposition
$\partial L/\partial t$	rotation rate	difference of consecutive observed pairs	measured
T_{NBI}	beam torque	no 0-D channel exists	absent (§6.7)
$T_{\mathrm{NTV}} \sim \delta B^2$	non-axisymmetric braking	Mirnov, decimated to 100 Hz	closed negative
$\nabla\cdot\Pi_\varphi$	turbulent momentum flux	BES sees the density fluctuation	**measured → null**
T_{int}	residual stress	needs ∇T_i; we hold a scalar	not reachable

4.2 Two conservation laws fix the input routing

The V_{rot} branch of `seq_v2` does not see the fast diagnostics. That is not a modelling preference; it is what reading the two transport equations term by term returns. Ion energy and toroidal angular momentum obey

$$\frac{\partial}{\partial t}\left(\tfrac{3}{2} n_i T_i\right) + \nabla\cdot\mathbf{q}_i = Q_{ei}\,(T_e - T_i) + S_i, \tag{3}$$

$$\frac{\partial}{\partial t}\left(n_i m_i \langle R^2\rangle \omega_\varphi\right) + \nabla\cdot\Pi_\varphi = T_{\mathrm{NBI}} + T_{\mathrm{NTV}} + T_{\mathrm{int}} + \cdots \tag{4}$$

and both constrain a *rate*, not a level. The question to ask about rotation is therefore not whether the fast diagnostics correlate with V_{rot} but whether they correlate with dV_{rot}/dt. Checking each term of (4) against the channels we hold gives Table 1.

T_{NTV} is already closed: Mirnov is decimated to 100 Hz, and amplitude proxies such as |MC| and a rolling RMS never helped (§6.7). The one term left was the turbulent momentum flux, reachable in principle because BES sees the density fluctuation it rides on. The measurement is null:

$$\underbrace{r\big(\mathrm{BES},\, dV_{\mathrm{rot}}/dt\big) = -0.006, \qquad r\big(\mathrm{ECEI},\, dV_{\mathrm{rot}}/dt\big) = -0.003}_{\text{momentum equation}} \quad \text{versus} \quad \underbrace{+0.070, \qquad +0.078}_{\text{energy equation } (dT_i/dt)}$$

The same asymmetry appears at the level of the quantities themselves (BES→T_i +0.341, ECEI→T_i +0.311 against V_{rot} +0.027 and +0.005). *Because the same measurement finds a coupling for the energy equation, the rotation null is not a limitation of the method.* With T_{NBI} absent from the dataset, T_{NTV} closed and T_{int} unreachable, the last term is null as well, so no route exists by which our inputs reach V_{rot} through (4). Splitting the branches at the encoder is the implementation of that audit, and it is why perturbing all 15 fast channels leaves the V_{rot} output bit-identical (§6.7). The remaining lever is acquisition, not modelling (§7).

4.3 The time scales the model sits on

Three times define this problem. First, the grid spacing is $\Delta t = 10$ ms, so the Nyquist frequency is

$$f_{\mathrm{Nyq}} = \frac{1}{2\,\Delta t} = 50 \text{ Hz}. \tag{5}$$

The mode rotation Mirnov would have to carry is in the kHz band—EHO harmonics reported for discharges in this dataset are near 4 and 8 kHz—and a dB/dt stream decimated to 100 Hz without an anti-aliasing filter cannot represent it, by the sampling theorem. The lag-1 autocorrelation of −0.009 against BES's +0.568 (§6.7) is the *confirmation* of that fact, not its basis.

Second, the time that couples electrons to ions. From the Braginskii electron collision time $\tau_e = 3.44\times10^5\, T_e^{3/2}/(Z\, n_e \ln\Lambda)$ (seconds, T_e in eV, n_e in cm^{-3}), the temperature equilibration time is

$$\tau_{\mathrm{eq}} \simeq \frac{m_i}{2m_e}\,\tau_e. \tag{6}$$

For deuterium with $\ln\Lambda = 17$ and $Z = 1$, τ_{eq} spans 8.3–305 ms over $T_e = 0.5$–3 keV and $n_e = 2$–5×10^{19} m^{-3}, and **8–59 ms** in the corner corresponding to our observed T_i median of 593 eV. The saturation point of

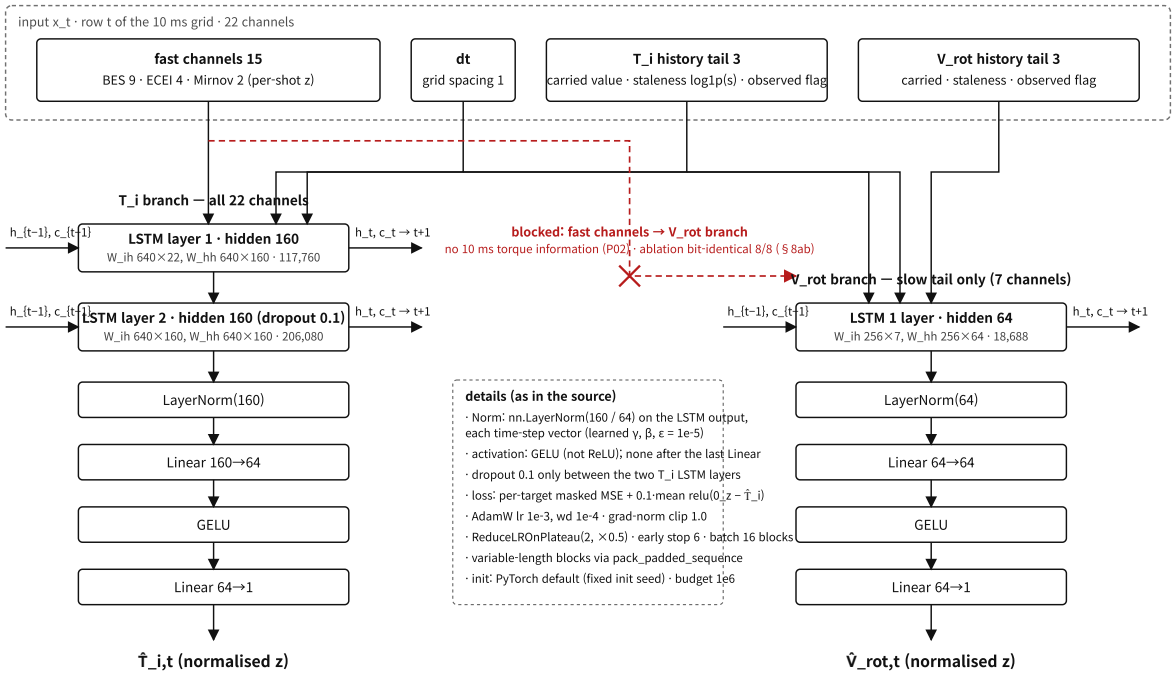


Figure 2: `seq_v2` for one grid row t. The 22 input channels (15 fast: BES, ECEI, Mirnov, per-shot z; Δt; two history tails of carried value, staleness and observed flag) feed two causal LSTM branches. The T_i branch reads all 22 channels (2 layers, 160 units; 117,760 + 206,080 parameters), the V_{rot} branch reads only the seven slow-tail channels (1 layer, 64 units; 18,688); each is followed by a per-step LayerNorm and a Linear–GELU–Linear head (no activation after the last layer). The dashed red path from the fast channels to the V_{rot} branch does not exist by construction; the recurrent state (h, c) passes from $t-1$ to t to $t+1$; the loss is the per-target masked MSE on rows with an actual CES measurement plus $0.1\,\mathrm{relu}(0_z - \hat{T}_i)$. 357,570 parameters in total.

contiguous causal context measured in this work is about 50 ms, and the two numbers are the same order. This is an order-of-magnitude consistency statement and nothing more: the plasma parameters in (6) are quoted rather than measured here (BES and ECEI are in instrument units, so this work does not hold n_e and T_e in physical units), and the prefactor carries an order-unity convention.

Third, the time over which a signal resembles itself. Autocorrelation $1/e$ times measured across the 641 files agree to within 10%: T_i 159 ms, BES 161 ms, ECEI 147 ms. V_{rot} does not: 54% of its observations are instrument repeats, so removing held values gives 16 ms and keeping them gives more than 300 ms, a factor of over 19. **The V_{rot} relaxation time cannot be measured in this dataset.** T_i does not suffer this, having one held row in 641 files. The held pathology costs not only 54% of the values but the ability to characterize the quantity's own dynamics, which makes it an acquisition task of its own (§7). These autocorrelations are computed without detrending and are dominated by the discharge envelope, so they are not quoted as transport times τ_E or τ_φ.

4.4 Full-grid causal sequence backbone

`seq_v2` runs two independent causal LSTMs over the 22-channel grid sequence of a segment (§3.2; Fig. 2, cell in Fig. 3). The T_i branch (2 layers, 160 units) reads the full state—fast diagnostics, both targets' carried values, staleness flags and Δt—and the V_{rot} branch (1 layer, 64 units) reads *only the seven non-fast channels*: Δt and the per-target carried value, staleness and observation flag. That wiring is the implementation of the

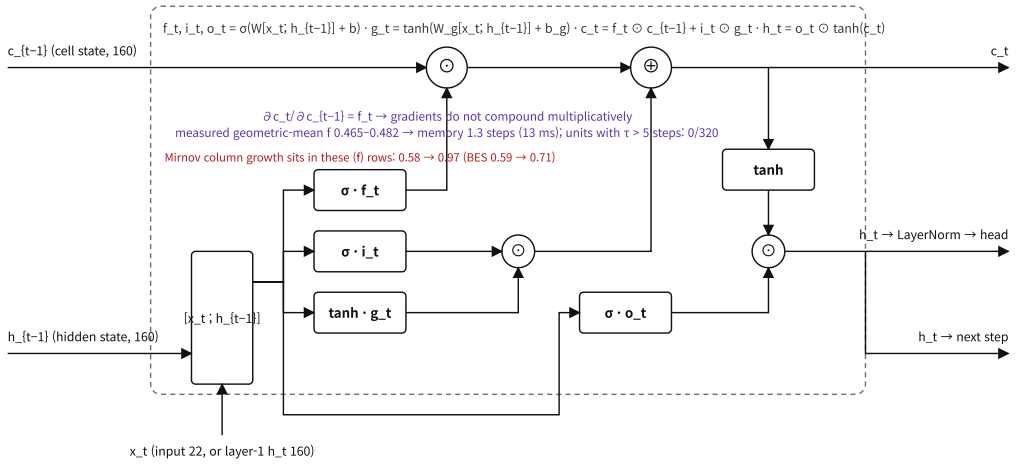


Figure 3: One LSTM cell (T_i branch, layer 1, 160 units). The cell state is multiplied by the forget gate and incremented by the gated candidate, $c_t = f_t \odot c_{t-1} + i_t \odot g_t$, so $\partial c_t/\partial c_{t-1} = f_t$ and the memory length is set by f rather than by compounded Jacobians. Purple: values measured on the frozen weights (geometric-mean f 0.465–0.482, memory 1.3 steps; 0 of 320 units with $\tau > 5$ steps). Red: where the Mirnov input columns grow during training (forget-gate rows, 0.58 → 0.97 against 0.59 → 0.71 for BES).

audit in §4.2. Each branch ends in LayerNorm and a small GELU head; outputs are normalized $[T_i, V_{\mathrm{rot}}]$ per row and the loss is (1) applied over every labelled row of the segment. Two consequences follow from the framing. First, the model's *reach* into the past is the whole segment rather than a fixed window—exactly the quantity that separates it from every windowed variant in §6.3. Second, the routing holds at the *encoder*: a shared recurrent state would carry fast-diagnostic information into the V_{rot} head no matter how the head is wired, whereas here perturbing all 15 fast channels leaves the V_{rot} output bit-identical (§6.7).

What the backbone computes. A pre-registered readout of the thirty frozen `seq_v2` checkpoints found that weight magnitude carries no stable information (first-layer column norms are flat across the fifteen fast channels, the largest weights sit on the least influential input, and per-channel ranks do not reproduce across splits), whereas the input–output response does: the T_i output is an exponentially weighted average of the past observed T_i plus a same-instant response to the fast channels, with a forget-gate memory of 1.3 grid steps. Scored on the frozen TEST rows, an unfitted running mean alone carries 73–82% (cut) of the backbone's skill over persistence, and a ridge-fitted linear term in the 15 ten-millisecond increments raises it to 77–85% (4/4 splits above the registered 0.75 bar). Trained from scratch as a 20-parameter equation under the identical protocol with that increment term, $\hat{y}_t = m_t + \sum_c a_c\,\kappa \tanh(\Delta x_{c,t}/\kappa) + b_0$ with an observation gain $K(s) = K_\infty + (K_0 - K_\infty)e^{-s/s_K}$ of the gap s since the previous measurement, the same physical numbers return on every cut split ($K_0 = 0.30$–0.31, $K_\infty = 0.52$–0.54, $s_K = 37$–40 ms, $\sum a_{\mathrm{BES}} = +43$ to $+51$, $\sum a_{\mathrm{ECEI}} = +39$ to $+46$ eV per σ, $\sum a_{\mathrm{MC}} \approx 0$), but the equation keeps only 50–66% of the backbone's PCHIP-relative skill (cut), below the registered 0.70 bar, and a single cross term adds less than 0.001. The Jacobian of the output on every frozen TEST row then showed that the form, not a missing state-dependent term, was the gap: the backbone's fast-channel response is a ≈100 ms high-pass (the current value minus the preceding ten-step mean) whose gain is flat to 2σ and switched off beyond 3σ, with no dependence on the recent increments. Re-trained with exactly that form and the same 20 parameters, $\hat{y}_t = m_t + \sum_c a_c\, d_{c,t}\, \sigma\big((\theta - |d_{c,t}|)/0.5\sigma\big) + b_0$ with $d_{c,t} = x_{c,t} - \overline{x}_{c,\,t-10..t-1}$, the equation keeps 72–78% of the backbone's PCHIP-relative skill and 87–90% of its persistence-relative skill on 4/4 cut splits, sits above the increment-form equation on every split by a paired $+0.034$ to $+0.042$ (CIs above zero), and learns $\theta = 2.9$–3.0σ, $\sum a_{\mathrm{BES}} = +48$ to $+53$, $\sum a_{\mathrm{ECEI}} = +44$ to $+50$ eV per σ, $K_0 = 0.27$–0.30. The remaining quarter of the PCHIP skill is the network's nonlinear event handling, which no low-order function of the

fast-channel history reproduces. These are measurements of the model; the headline numbers remain the backbone's.

4.5 The windowed control: observation-masked attention pooling

The control encodes each modality (BES, ECEI, MC) with its own time-aware 1-D CNN over the W=2 window, reads the CES history with a bidirectional GRU (hidden 64) whose per-timestep outputs are pooled by two independent multi-head additive attention readouts, one per target, *hard-masked* so that attention mass can land only on rows where *that* target was genuinely measured—interpolation's own inductive bias (use only observed samples) written into attention at zero parameter cost. Its T_i head fuses fast diagnostics, time and history; its V_{rot} head fuses history and time only. Everything the two models share—the data contract, hold-free treatment, per-target masking, the split, the scored population—is pinned identically, so their comparison is row-for-row paired.

4.6 The interpretable rung

`b3k8` (§6.8) keeps `seq_v2`'s framing and routing but replaces each head by an *exact* decomposition, $\hat{y} =$ carried value $+ \sum_k w_k z_k + b$, where $z \in [-1, 1]^K$ is a tanh latent from a small causal GRU (64/32 units, K=8 for T_i, 4 for V_{rot}) and the readout is zero-initialized so training starts exactly at persistence. It has 21,498 parameters, 6% of the backbone.

4.7 The optimization setup, and the state of its justification

Training uses AdamW ($\eta = 10^{-3}$, weight decay 10^{-4}), batch 16 segments, gradient norm clipping at 1.0, `ReduceLROnPlateau` on validation masked MSE (patience 2, factor 0.5) and early stopping (patience 6, cap 30 epochs; the confirmatory runs stopped at 14–25 epochs).

These values are *inherited* from the windowed pipeline rather than derived, and this work does not hide that. The data and structure axes were all closed by controlled experiments—window size (§6.14), held treatment (§3.4), the two populations (§3.5), complexity and width (§6.8)—but the optimization axis never received one. A single, and indirect, piece of evidence defends the setting: the *budget-equalization* condition of the backbone gate trains both models for a fixed 10 epochs with validation-based checkpoint selection forbidden, and even there the backbone's paired T_i gain is positive on all four splits (+0.063, +0.033, +0.045, +0.030; §6.3). The conclusion is therefore not an artifact of the training schedule.

The loss axis has since received one controlled experiment (§6.10): a Huber loss with its scale pinned to the target's 10 ms reproducibility bound cut the median error by 8–10% in the cut population while holding the aggregate skill, and the backbone's loss stays MSE by the pre-registered rule. That experiment also taught a coupling: selecting an arm that changes the objective on val MSE picks a nearly untrained checkpoint (epoch 2–4) in the tail-dominated inclusive population. The objective and the checkpoint-selection metric are a pair, and when one is changed, how the other was set is part of the result.

"Why batch 16" now has a measured answer. The gradient noise scale $B_{\mathrm{simple}} = \operatorname{tr}\Sigma / \|G\|^2$ is the batch size at which a gradient's signal and its noise are comparable, and taking the sampling unit to be the block—what the training loop actually draws—it can be computed exactly on the training set rather than estimated. With g_i the gradient of block i's own masked loss,

$$G = \tfrac{1}{N}\sum_i g_i, \qquad \operatorname{tr}\Sigma = \tfrac{1}{N}\sum_i \|g_i\|^2 - \|G\|^2, \qquad B_{\mathrm{simple}} = \frac{\operatorname{tr}\Sigma}{\|G\|^2}. \tag{7}$$

Evaluated over all 452 training blocks at seven points during training, the post-initial median is **33.6 blocks** (geometric mean 28.4, range 6.0–61.4). **Batch 16 sits at about half the measured noise scale.** Below B_{simple} batch size and step count trade roughly linearly, so 16 is neither wasteful nor obviously suboptimal; doubling it to 32 would roughly halve the step count at the same data efficiency, and going far above 60 would begin to waste examples.

This is an order-of-magnitude statement. $\frac{1}{N}\sum_i \|g_i\|^2$ and $\operatorname{tr}\Sigma$ fall monotonically after the first epoch (10.5 → 7.5 and 10.3 → 7.3), but $\|G\|^2$ moves between 0.15 and 1.74, and since it is computed exactly over all 452 blocks that motion is a real property of where the Adam trajectory sits at an epoch boundary rather

than measurement error. What is stable is the noise share: after the first epoch **86–98% of a single block's gradient energy is noise**, leaving 2–14% of signal. The measurement is made with dropout disabled, so as to capture the noise from sampling examples alone; the training loop's own effective noise scale is therefore *at least* the figure above, and the figure concerns the compute-versus-steps trade rather than final skill.

The second measurement, μP—a parameterization under which the optimal learning rate does not shift with width—has not been run. Adopting it would make the distinction between "capacity does not help" and "the learning rate did not match that width" unnecessary by construction for the width sweep of §6.8.

What the search taught. Across two rounds of controlled architecture iterations on the windowed family (selection on clean validation skill only), attention pooling was the one mechanism that survived, and capacity scaling, skip topologies and extra extractors never helped. Under the confirmed protocol the same lesson recurs twice more: observation-masked causal attention added to `seq_v2` is a small, consistently positive but non-significant gain (§6.3), and a five-point width sweep of `seq_v2` is flat (§6.8). The bottleneck in this problem is information, not capacity, and (2) and Table 1 say where that bottleneck sits for each target.

5 Evaluation methodology

Metric. All errors are denormalized to physical CES units and computed per target. The primary score is Murphy's skill [14] against the pre-registered headline baseline, $\text{skill}_{\text{PCHIP}} = 1 - \text{MSE}_{\text{model}}/\text{MSE}_{\text{PCHIP}}$ (positive = model better; 0 = tie). Every arm (model and all baselines) is scored on the *identical* (file, row) sample set under the same per-target keep mask, verified bit-identically on the population keys before any paired comparison. Interpolants exclude the target's own value (no leakage). Neither the interpolants nor a model's input bridges a segment boundary; where a neighbour beyond the boundary would be needed the interpolant predicts the persistence value instead (no population shrinkage; PR2).

Baseline ladder. Persistence and a local past-only AR are the causal references of the original pre-registration; linear, PCHIP and a Gaussian process (exact Matérn-3/2 + white noise, local nearest-16+16 fit, per-sample marginal-likelihood grid selection, inheriting the PR2 fallback) read the future. To the causal side we add a **causal GP**: the same GP restricted to the 16 past neighbours, with the same NaN conditions so the scored population does not move. It is decisively the strongest deployable baseline (seed-42 T_i RMSE 164.3 against persistence's 197.2, cut population), and "beats every deployable causal method" is judged against it, not against persistence.

Three-way split; selection never sees TEST. Files are split train/val/test; the test files of every split were reserved *before* any architecture search and are never read during model selection. The canonical (seed-42) test set contains $n = 32{,}589$ observed T_i rows across 96 shots and $n = 10{,}463$ genuine V_{rot} rows across 60 shots in the cut population (32,721 / 10,461 inclusive); the four splits (seeds 42, 1, 7, 123) hold 32.6–35.9k T_i rows over 96 shots and 10.5–14.5k V_{rot} rows over 60–66 shots each. Every number in this paper is read from frozen run directories by a single collection script, so the text, the tables and the figures cannot drift apart.

Pre-registration. "Pre-registered" has one meaning in this paper: the hypotheses, decision rules and numerical predictions of a batch were committed to the project repository before that batch was run, with the commit recorded; there is no third-party registry. Rules fixed in writing before the corresponding test numbers were viewed: **PR1** the headline comparator is PCHIP (chosen a priori for monotonicity/no ringing at transients; the full ladder is also reported); **PR2** interpolation predicts everywhere the model is scored, with persistence fallback where no future neighbour exists—and the fallback rate is reported (0.3–0.4% of scored T_i rows, but **40–44% of scored V_{rot} rows**, so "vs. PCHIP" for V_{rot} is two-fifths "vs. persistence"); **PR3** a minimum test-set floor ($\geq$15 shots, $\geq$3,000 observed T_i samples; met everywhere); **PR4** significance means a shot-clustered bootstrap 95% CI on the skill excluding zero. To these the confirmed protocol adds: hold-free training and scoring; W=2 by the plateau-minimal rule of a 24-run sweep (§6.14); a per-file sample cap of 500 in the windowed family; the two co-primary populations of §3.5; and *model-decision rules committed before*

TEST scoring for the backbone gate, the one attention candidate, the interpretable rung and the width sweep (§6.3). Threshold sensitivity (2.5/3/4 keV) is reported.

Shot-clustered paired bootstrap. Adjacent CES rows within a discharge are strongly correlated; treating samples as independent would grossly understate uncertainty. Per-sample paired errors $\mathrm{SE}_{\mathrm{model}} - \mathrm{SE}_{\mathrm{PCHIP}}$ are aggregated by shot, and whole shots are resampled with replacement (10,000 resamples, fixed seed). The resulting CI reflects genuine shot-to-shot generalization; its cost is that the effective sample size is the number of shots (≈96 for T_i, 60–66 for V_{rot}), which bounds statistical power throughout. Model-vs-model comparisons (backbone vs. control, rung vs. backbone) use the same paired bootstrap on the same rows.

Replication beyond the four splits. A count of passing splits is not a test (4/4 by chance alone is 1/16), and the four frozen splits share most of their TRAIN discharges, so an interval over them is too narrow. Two pre-registered replicates therefore re-ran the backbone as shot-level repeated 5-fold cross-validation on the 639-discharge universe (TEST 128, VAL 128, TRAIN 383 discharges per fold; every discharge tested once per repeat; ten repeats, eight of them run after the pooled readout rule was registered; 100 GPU runs in the headline's environment). Two readouts were registered before any number. (i) *Fold-level*: the mean of the fold skills with the Nadeau–Bengio overlap-corrected t-interval, $\bar{S} \pm t_{0.975,\,n-1}\sqrt{(1/n + n_{\mathrm{test}}/n_{\mathrm{train}})\,s^2}$ [55], which widens the plain interval by about 2× at $n_{\mathrm{test}}/n_{\mathrm{train}} = 0.33$ and does not shrink to zero with more folds [56]. (ii) *Pooled*: each repeat's five TEST folds concatenated so that every discharge is scored once in one ratio skill, with a two-level bootstrap that draws one repeat (training-partition variation) and then resamples its discharges (shot variation)—a hierarchical bootstrap in the sense of Saravanan et al. [57] and Ren et al. [58] with the training repeat as the outer level, with the structure of the Multi-Bootstrap [59] and of the decadal-prediction skill-score bootstrap [60], assembled from standard pieces and not claimed as new; a cell passes iff the 2.5th percentile exceeds zero. Repeats and discharges are resampled as crossed factors (every discharge is scored in every repeat); bootstraps of crossed arrays are not exact, their variance estimate is biased [80] toward overestimation [81], so the interval is conservative. For T_i the two agree to the second decimal. For V_{rot}, whose ≈80-discharge folds are set by a few heavy-error discharges (fold-to-fold s.d. 0.18), the fold-level interval spans zero while the pooled one does not; the registration named the pooled two-level readout, computed on the eight repeats run after the rule was written, as the deciding one for the replicate, and Table 4 reports both.

The MNAR problem, and what we do about it. Truly missing points have no ground truth, so skill is measured by masking *observed* points. Because CES missingness is MNAR (drop-out at low S/N, ELMs, transitions), observed-point skill is an optimistic estimate of performance at genuinely missing points. §6.5 reweights the scored points onto the covariate distribution of the *actually missing* ones and reports how much of the skill survives, together with what fraction of the missing population is in the analysis' domain at all.

6 Results

6.1 The nowcaster has the lowest RMSE of every causal baseline

Table 2 shows the physical-unit RMSE ladder on the canonical test split. `seq_v2` has the lowest RMSE for both targets among all methods that do not read the future, 4% (T_i) and 18% (V_{rot}) below the causal GP—the strongest deployable arm—and below the future-using interpolants; the offline GP, which averages noisy anchors rather than passing through them, is the one arm with a lower T_i RMSE (153.8 vs. 157.8; paired skill −0.05 to +0.11 with the CI including zero on 7/8 cells, §6.2). This is the most robust result in the paper: for any online setting, where future CES is by definition unavailable, the sequence nowcaster has the lowest error.

Table 2: Physical-unit RMSE ladder, canonical held-out test split (seed 42), cut population, genuine measurements only (n = 32,589 T_i, 10,463 $V_{\rm rot}$). Lower is better. The three light rows read the future. Inclusive population: `seq_v2` 363.0 / 23.7, PCHIP 412.4 / 30.2, causal GP 394.6 / 28.8, persistence 478.0 / 33.4 (T_i / $V_{\rm rot}$); the fit-failure spikes more than double every T_i RMSE without changing the order.

Arm	Information access	T_i RMSE (eV)	$V_{\rm rot}$ RMSE (km/s)
`seq_v2` (nowcaster)	fast diag. + past CES, whole segment	**157.8**	**23.6**
Windowed control (W=2)	fast diag. + 2 rows of past CES	169.2	26.1
Causal GP	past 16 CES neighbours	164.3	28.8
Persistence	last observed CES	197.2	33.4
AR (local, causal)	past CES only	472.2	51.0
GP (offline)	past + future CES	153.8	24.7
Linear interpolation	past + future CES	169.8	29.0
PCHIP interpolation	past + future CES	173.6	30.2

Table 3: Held-out test skill$_{\rm PCHIP}$ of `seq_v2` with shot-clustered 95% CIs (10,000 resamples), both populations; the last column is the paired skill against the causal GP on the same rows. Bold = PASS (CI excludes 0).

Target	split	cut: skill$_{\rm PCHIP}$ (95% CI)	inclusive: skill$_{\rm PCHIP}$ (95% CI)	vs. causal GP (cut / incl.)
T_i	42	**+0.174** [+0.097, +0.236]	**+0.225** [+0.109, +0.293]	**+0.078** / **+0.154**
	1	**+0.248** [+0.188, +0.295]	**+0.238** [+0.153, +0.302]	**+0.133** / **+0.169**
	7	**+0.257** [+0.199, +0.302]	**+0.292** [+0.232, +0.344]	**+0.138** / **+0.123**
	123	**+0.264** [+0.188, +0.320]	**+0.316** [+0.186, +0.392]	**+0.105** / **+0.149**
$V_{\rm rot}$	42	**+0.390** [+0.077, +0.591]	**+0.384** [+0.066, +0.586]	**+0.331** / **+0.324**
	1	+0.183 [−0.028, +0.280]	**+0.195** [+0.000, +0.286]	+0.130 / **+0.143**
	7	+0.135 [−0.358, +0.269]	+0.132 [−0.362, +0.266]	+0.020 / +0.018
	123	+0.305 [−0.049, +0.437]	+0.304 [−0.065, +0.433]	**+0.134** / +0.132

6.2 Headline: T_i skill over future-using interpolation excludes zero on four independent splits, in both populations

Table 3 reports the backbone on the four independent held-out splits (seeds 42, 1, 7, 123—the latter three never used in any selection step) in the two co-primary populations, against PCHIP and against the causal GP; Fig. 4 shows the PCHIP intervals.

- **T_i: PASS on all four splits, in both populations.** skill$_{\rm PCHIP}$ = +0.17 to +0.26 (cut) and +0.23 to +0.32 (inclusive); every shot-clustered CI excludes zero, and in every one of the eight population–split cells the skill over the *causal GP* (+0.08 to +0.17) and over persistence (+0.36 to +0.46) also has its CI above zero. This is the unqualified claim of the paper. Against the offline GP the backbone's paired skill is −0.05 to +0.11 with the CI including zero on 7/8 cells, so the observed-population claim reads "skill above zero over the pre-registered interpolants and every deployable causal method, and an interval including zero against a future-using, per-sample-tuned smoother". The replicate (Table 4) reproduces every verdict: vs. PCHIP the overlap-corrected fold-level interval is [+0.19, +0.25] (cut) and [+0.19, +0.31] (inclusive), the pooled two-level interval [+0.19, +0.25] and [+0.19, +0.29], and 96/100 folds pass on their own.

- **$V_{\rm rot}$: intervals including zero on the four-split count, a registered interval above zero on the replicate.** Point estimates are positive on every split in both populations, but PR4 passes on 1/4 (cut) and 2/4 (inclusive)—seed 42 in both, seed 1 at [+0.000, +0.286] inclusive; against persistence $V_{\rm rot}$ passes on 3/4 splits in both populations (+0.30 to +0.50; seed 7's CI touches zero) and against the causal GP on 2/4. One or two significant splits out of four is what noise would produce, so the four-split count alone supports no rotation claim. In the replicate, scored discharge by discharge across eight fresh partitions, the $V_{\rm rot}$ skill over PCHIP is [+0.07, +0.70] (cut) and [+0.05, +0.68] (inclusive), over persistence [+0.28, +0.76] and [+0.14, +0.75], and over the causal GP [+0.05, +0.70] and [+0.02, +0.68]

Table 4: Pre-registered replicate of the headline as shot-level repeated 5-fold cross-validation (ten repeats, every discharge tested once per repeat, 100 runs). Fold-level: Nadeau–Bengio overlap-corrected 95% t-interval on the mean of the 50 fold skills, with the number of folds whose own shot-clustered CI excludes zero. Pooled: two-level (one repeat, then its discharges) bootstrap 95% interval on the per-repeat pooled skill over the eight repeats run after the rule was registered. Bold = interval excludes 0.

Target	vs.	cut: fold-level (pass)	cut: pooled	inclusive: fold-level (pass)	inclusive: pooled
T_i	PCHIP	$[+\mathbf{0.186}, +\mathbf{0.253}]$ (50/50)	$[+\mathbf{0.188}, +\mathbf{0.248}]$	$[+\mathbf{0.188}, +\mathbf{0.307}]$ (46/50)	$[+\mathbf{0.192}, +\mathbf{0.294}]$
	persistence	$[+\mathbf{0.368}, +\mathbf{0.410}]$ (50/50)	$[+\mathbf{0.369}, +\mathbf{0.408}]$	$[+\mathbf{0.361}, +\mathbf{0.445}]$ (50/50)	$[+\mathbf{0.364}, +\mathbf{0.438}]$
	causal GP	$[+\mathbf{0.084}, +\mathbf{0.138}]$ (47/50)	$[+\mathbf{0.085}, +\mathbf{0.132}]$	$[+\mathbf{0.067}, +\mathbf{0.170}]$ (40/50)	$[+\mathbf{0.076}, +\mathbf{0.157}]$
	offline GP	$[-0.044, +0.020]$ (1/50)	$[-0.043, +0.014]$	$[-0.026, +0.094]$ (12/50)	$[-0.017, +0.075]$
V_{rot}	PCHIP	$[-0.021, +0.405]$ (15/50)	$[+\mathbf{0.072}, +\mathbf{0.701}]$	$[-0.053, +0.391]$ (11/50)	$[+\mathbf{0.046}, +\mathbf{0.683}]$
	persistence	$[+\mathbf{0.190}, +\mathbf{0.601}]$ (42/50)	$[+\mathbf{0.277}, +\mathbf{0.763}]$	$[+\mathbf{0.159}, +\mathbf{0.587}]$ (38/50)	$[+\mathbf{0.139}, +\mathbf{0.747}]$
	causal GP	$[-0.060, +0.512]$ (11/50)	$[+\mathbf{0.045}, +\mathbf{0.701}]$	$[-0.096, +0.502]$ (10/50)	$[+\mathbf{0.023}, +\mathbf{0.683}]$
	offline GP	$[-0.234, +0.312]$ (0/50)	$[-0.119, +0.635]$	$[-0.268, +0.300]$ (0/50)	$[-0.139, +0.609]$

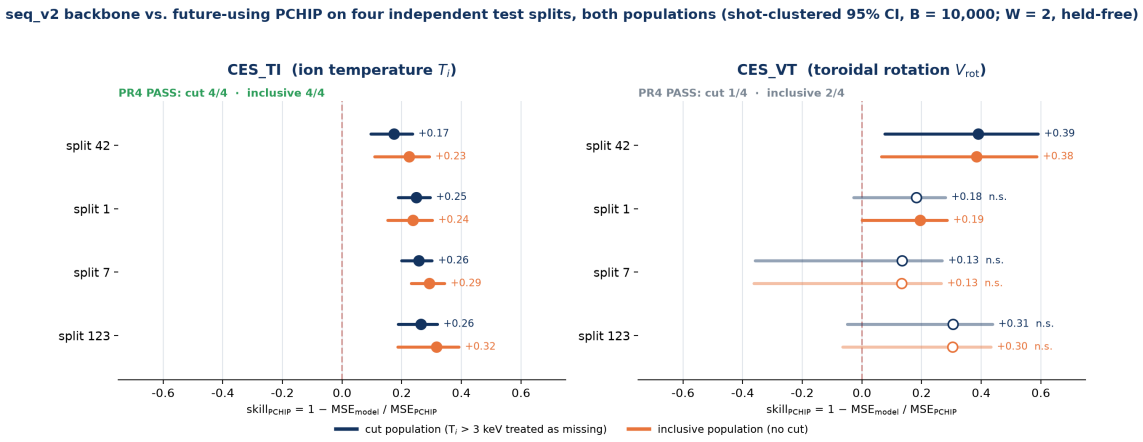


Figure 4: Forest plot of `seq_v2`'s skill$_{\text{PCHIP}}$ with shot-clustered 95% CIs across four independent held-out splits, both populations. T_i excludes zero on all eight cells; V_{rot} on three.

under the registered two-level bootstrap, every interval above zero, every one of the eight new model sets positive (+0.28 to +0.59 cut, +0.09 to +0.56 inclusive vs. PCHIP), while the fold-level overlap-corrected interval still spans zero ($[-0.02, +0.41]$ cut) because an 80-discharge fold's ratio is set by a few heavy-error discharges. The effect is a mean-squared-error, tail-carried one (§6.10); in the median the V_{rot} error ratio to persistence is 0.94–1.00, with the interval including 1 on 3/4 splits, at the 10 ms photon floor (§6.11), and the two statements are made together.

- **Why the inclusive numbers are higher.** Every arm scores *better* against PCHIP without the cut (backbone mean +0.268 vs. +0.236; control +0.238 vs. +0.173), because the spikes poison the interpolation anchors: a learned model discounts a spiked carried value where an interpolant cannot. §6.7 quantifies this component; it is the reason neither population is quoted alone.

6.3 The full-grid framing against the windowed control: the backbone gate

The sequence framing was adopted under a pre-registered four-condition gate, scored once in the cut population on a 4×4 grid of split seed × initialisation seed, each run paired row-for-row against its split's W=2 windowed control. The paired T_i skill is positive on **16/16 runs** (CI above zero on 13/16), with pooled mean $+\mathbf{0.081}$ and run-cluster 95% CI $[+0.067, +0.096]$, excluding zero; a budget-equalized arm (fixed 10 epochs, final weights, no validation selection) keeps the sign on 4/4 splits (+0.063, +0.033, +0.045, +0.030), and no run has

Table 5: Gap-stratified skill, four test splits pooled, shot-clustered 95% CIs, `seq_v2`. Bold = CI excludes zero. Rows with fewer than 50 samples are not scored.

Δt	n (cut / incl.)	shots	cut: vs. PCHIP	inclusive: vs. PCHIP
T_i				
≤ 15 ms	134,546 / 135,317	301	**+0.239** [+0.197, +0.274]	**+0.299** [+0.244, +0.347]
> 15 ms	3,422 / 3,334	265 / 263	**+0.268** [+0.187, +0.337]	**+0.206** [+0.108, +0.290]
> 45 ms	460 / 429	104 / 101	**+0.267** [+0.092, +0.414]	−0.004 [−0.304, +0.246]
V_{rot}				
≤ 15 ms	51,689	197	**+0.233** [+0.020, +0.318]	**+0.233** [+0.020, +0.317]
> 15 ms	466 / 456	130	**+0.418** [+0.104, +0.680]	**+0.432** [+0.128, +0.696]
> 45 ms	14	7	—	—

a significant V_{rot} deficit, so all four conditions hold. What the framing buys is reach: the windowed control's skill over the causal GP is +0.041 with the CI including zero on 3/4 splits (cut), and the backbone's is +0.08 to +0.17 with CIs above zero on 4/4 in both populations. Every model decision was taken on validation data or under a rule committed in writing before the corresponding TEST scoring, with TEST scored once per decision. Under the same rule the one architectural candidate explored afterwards—observation-masked causal attention over each target's own past observation steps, added to `seq_v2`—has a paired T_i skill against the backbone of +0.009, +0.013, +0.033, +0.020, with the CI above zero on 1/4 splits against a pre-committed bar of $\geq 3/4$, so it was not promoted.

6.4 Where the skill lives: gap regime

Stratifying by Δt, the time since the last observed CES value, ≥96% of scored targets lie in the adjacent regime ($\Delta t \leq 15$ ms); per split the wide bins hold only hundreds of samples, so we *pool the four test splits* and cluster the bootstrap on the physical discharge (Table 5). As Δt grows PCHIP's task gets *easier*—it interpolates between two real observations that straddle the gap—while ours gets harder, so this is the regime where a causal method is least expected to win.

The T_i skill is not confined to adjacent history. Beyond 15 ms the backbone's skill over future-using PCHIP is +0.268 and +0.206 in the two populations (both CIs above zero) and its skill over persistence there is +0.40 and +0.43 (CIs [+0.34, +0.45] and [+0.35, +0.52]). Beyond 45 ms the skill over PCHIP is +0.267 [+0.09, +0.41] under the cut and −0.004 with CI [−0.30, +0.25] in the inclusive population: 429 rows over 101 shots is exactly where a handful of spike-anchored rows can dominate a stratum, and we report the included-zero interval as such.

V_{rot} skill over interpolation excludes zero in the one place it is hardest for interpolation. The global V_{rot} interval that includes zero is the adjacent regime's; over $\Delta t > 15$ ms the backbone's skill over PCHIP is +0.418 [+0.10, +0.68] and +0.432 [+0.13, +0.70] in the two populations (130 shots)—the single unconditional V_{rot} interval above zero in the paper—and its skill over persistence at ≤ 15 ms is +0.39 [+0.21, +0.45]. The widest bin has 14 rows and is not scored.

6.5 How much of this survives at the points that are actually missing?

Everything above is measured on CES points that happen to be *observed*. We can never measure error where there is no ground truth—but we can measure how skill varies over covariates that are computable at observed *and* missing rows, and reweight. Two covariates carry the mechanism: Δt since the last genuine observation, binned at 15/25/45 ms, and the input-only local-activity flag of §6.9, which is defined at a missing row because it is computed from a neighbour set that excludes the row itself. We post-stratify on $\Delta t\times$ activity, drop any stratum with fewer than 30 scored samples, and report the surviving coverage next to every estimate; the weight grid is population-consistent (the cut is applied to it too). The assumption—that within a stratum missing and observed rows are exchangeable—is stated, not assumed away.

Table 6: T_i skill of `seq_v2` reweighted from the observed population onto the genuinely-missing, in-domain one (shot-clustered 95% CIs at fixed stratum weights). Bold = CI excludes zero; *unweighted* is the headline of Table 3.

		vs. PCHIP		vs. persistence	
Population	split	unweighted	missing-matched (95% CI)	unweighted	missing-matched (95% CI)
cut	42	+0.174	+0.140 [−0.071, +0.290]	+0.360	**+0.398** [+0.278, +0.518]
	1	+0.248	**+0.164** [+0.062, +0.250]	+0.406	**+0.366** [+0.299, +0.448]
	7	+0.257	+0.203 [−0.024, +0.346]	+0.403	**+0.310** [+0.227, +0.398]
	123	+0.264	**+0.283** [+0.069, +0.392]	+0.415	**+0.381** [+0.246, +0.464]
inclusive	42	+0.225	**+0.140** [+0.033, +0.250]	+0.423	**+0.443** [+0.141, +0.623]
	1	+0.238	**+0.217** [+0.086, +0.319]	+0.436	**+0.383** [+0.273, +0.536]
	7	+0.292	**+0.167** [+0.067, +0.265]	+0.406	**+0.283** [+0.172, +0.380]
	123	+0.316	**+0.221** [+0.050, +0.320]	+0.459	**+0.337** [+0.164, +0.455]

Table 7: Temporal (campaign) split, four initialisations, TEST block of 97 later discharges. Entries are skill$_{\text{PCHIP}}$; **bold** = PR4 PASS. The last column is the paired T_i skill of `seq_v2` against the windowed control on the same rows.

Population	arm	T_i vs. PCHIP (init 42 / 1 / 7 / 123)	PASS	T_i vs. causal GP	`seq_v2` − control
cut	windowed control (OFF)	+0.027 / **+0.091** / −0.001 / **+0.061**	2/4	0/4	—
	control, per-shot std. (ON)	**+0.103** / **+0.107** / **+0.094** / **+0.107**	4/4	—	—
	`seq_v2`	**+0.187** / **+0.174** / **+0.181** / **+0.177**	4/4	4/4 (+0.11 to +0.12)	**+0.164** / **+0.091** / **+0.182** / **+0.124**
inclusive	windowed control (OFF)	+0.014 / +0.047 / +0.055 / +0.089	0/4	0/4	—
	`seq_v2`	**+0.173** / **+0.202** / **+0.198** / **+0.184**	4/4	4/4 (+0.13 to +0.16)	**+0.161** / **+0.163** / **+0.151** / **+0.104**

First, a scope result. The analysis' domain is "a genuine observation of that target within two rows", the W=2 criterion. Of the genuinely missing T_i rows, **54–68% are in-domain** per split; of the genuinely missing V_{rot} rows, **only 4–6%** are, because V_{rot} gaps are long blocks (coverage of surviving strata 0.99 for T_i, 0.73–0.76 for V_{rot}). The reweighted V_{rot} row therefore answers a question about one-twentieth of the missing mass and we draw nothing from it.

Second, the two comparisons come apart—and the split is the useful thing. Against **persistence—the baseline an online system actually runs—the T_i skill survives reweighting on all four splits in both populations** (+0.28 to +0.44, every CI excluding zero); the MNAR correction costs at most 0.12 of skill and on some splits nothing. Against **future-using PCHIP** the reweighted point estimates stay at +0.14 to +0.28 but the wide fixed-weight CIs pass through zero on two cut-population splits (2/4 significant; 4/4 inclusive). Missing points sit at larger Δt, which is where a two-sided anchor helps most and where the reweighting bootstrap is thinnest. The claim the evidence supports at the points that matter is therefore: *at genuinely missing, in-domain timesteps the nowcaster is significantly better than any causal CES-only method in both populations, and better than offline interpolation with a population-conditional significance (2/4 cut, 4/4 inclusive).* The windowed control behaves the same way (2/4 and 4/4 vs. PCHIP; 4/4 vs. persistence).

6.6 Does it survive a campaign shift?

The second thing deployment does that a random split does not is move forward in time. We order all 641 discharges by shot number (30801–32751) and cut strictly in time: **train 416 shots [30801, 31991], val 128 [32002, 32310], test 97 [32312, 32751]**, no test shot preceding any train shot. Everything else is pinned to the headline protocol, so the only changed variable is the split rule; the split is fixed, so the four runs vary the *initialisation* seed only, and we say so rather than presenting them as four splits. Three arms per population: the windowed control as trained for the headline (train-file-only normalization), the same control with per-shot standardization of the fast diagnostics (cut population), and `seq_v2` (Table 7).

Three things in that table are worth stating explicitly. First, the *windowed* model's skill over offline interpolation does not survive the shift: −0.001 to +0.091, with the CI above zero on 2/4 (cut) and 0/4

Table 8: Evaluation-time modality ablation of the W=2 windowed control, TEST, both populations: $\text{skill}_{\text{PCHIP}}$ (4-split mean) and the paired change against the un-ablated run (splits 42 / 1 / 7 / 123; * = significantly worse).

Inputs	target	cut	inclusive
Full (history + fast + time)	T_i	+0.173	+0.238
History + time only (*no fast*)	T_i	−0.125; paired −0.25* / −0.38* / −0.42* / −0.43*	+0.201; paired −0.03* / −0.04 / −0.03 / −0.09*
Fast + time only (*no history*)	T_i	−2.11; paired −4.6* / −1.8* / −2.3* / −1.9*	−1.16; paired −1.5* / −3.4* / −1.2* / −1.1*
Full	V_{rot}	+0.213	+0.206
History + time only (*no fast*)	V_{rot}	+0.213; paired +0.000 ×4 (**bit-identical**)	+0.206; paired +0.000 ×4 (**bit-identical**)
Fast + time only (*no history*)	V_{rot}	−2.89; paired −5.4* / −6.3* / −1.8* / −2.3*	−3.51; paired −6.9* / −7.8* / −1.9* / −2.2*

(inclusive) initialisations, and 0/4 against the causal GP. Second, its named repair works: standardizing each discharge's fast diagnostics by *its own* mean and variance at load time (targets untouched) takes the cut-population gate from 2/4 to 4/4, paired +0.078, +0.018, +0.095, +0.049 (3/4 significant), and leaves V_{rot} unchanged (−0.003 to +0.008)—the expected negative control, since the V_{rot} head never sees the fast diagnostics. Third, and this is the result of the section: **the sequence nowcaster does not collapse**. Its skill over PCHIP is +0.17 to +0.20 with the CI above zero on 4/4 initialisations in both populations, its skill over the causal GP +0.11 to +0.16 (CI above zero on 4/4 in both), and its paired skill over the windowed control +0.09 to +0.18 (CI above zero on 8/8); its V_{rot} skill over persistence is +0.31 to +0.34 with the CI above zero on 4/4 in both populations where the windowed control's interval includes zero on 4/4, and its V_{rot} margin over the control is +0.06 to +0.18 with the CI above zero on 8/8.

Why the control degrades—measured, not asserted. Between the train and test periods the median location shift, in the units the control was normalized in, is **1.22 σ for BES** and **0.53 σ for ECEI** (scale ratios 0.75 and 0.62) against **0.115 σ** (scale ratio 1.06) for the CES targets, a 5–11× larger drift, so train-file-only normalization does not transfer across the campaign boundary while per-shot standardization, which `seq_v2` uses by definition and which reads only that discharge's own data, does.

Two stress tests, one reading. The skill over the baselines a real-time system can actually run survives both stress tests in both populations with CIs above zero (+0.28 to +0.44 over persistence under reweighting, +0.11 to +0.16 over the causal GP under the campaign split). The skill over *future-using* interpolation survives the temporal shift in both populations (+0.17 to +0.20, CIs above zero) and the MNAR reweighting with population-conditional significance (+0.14 to +0.28, CI above zero on 2/4 cut and 4/4 inclusive).

6.7 The $T_i \leftrightarrow V_{\text{rot}}$ information asymmetry

A controlled input-modality ablation on the windowed control—one modality group zeroed *at evaluation*, no retraining, persistence and interpolants computed from the real history—explains *why* the two targets behave differently and re-grounds the routing of §4 under the confirmed protocol (Table 8).

- **History is indispensable for both targets.** Without the CES history the windowed model falls to −1 to −4 against PCHIP on both targets in both populations: neither target is predictable from 100 Hz diagnostics without its own anchor.
- **T_i: the margin over interpolation is fast-diagnostic information—in the cut population.** With BES/ECEI/MC zeroed the control drops *below* the interpolator (−0.10 to −0.18; paired −0.25 to −0.43, 4/4 significant). The physical channel is collisional electron–ion coupling ($t_{ei} \propto T_e^{3/2}/n_e$): ECEI supplies T_e and BES supplies n_e structure.
- **In the inclusive population the same history-only model still has skill +0.15 to +0.23 over PCHIP**, and the fast channels add only 0.03–0.09 (2/4 significant): the interpolant's anchors are spiked and a learned model discounts them where an interpolant cannot. The spike-inclusive margin therefore contains a *spike-robustness* component that is not fast-diagnostic information, which is the measured reason this paper reports two populations rather than the more flattering one alone; the cut population is the one that isolates what the fast diagnostics contribute.

- $V_{\rm rot}$**: information is entirely CES history, by construction and by measurement.** Zeroing the fast diagnostics leaves the $V_{\rm rot}$ output *bit-identical* on 8/8 population–split cells (the routing is structural), and `seq_v2`'s and `b3k8`'s $V_{\rm rot}$ branches pass the same perturbation test. Toroidal rotation is governed by momentum sources none of our inputs observe—the external NBI torque and intrinsic rotation drives [82]—and the Mirnov signals are sampled at 100 Hz, aliasing away the kHz mode-rotation frequencies that could otherwise proxy rotation (§7).

This asymmetry was predicted from physics before the ablation and is confirmed by it; the $V_{\rm rot}$ null is a finding about diagnostic information content, not a model failure. It also motivates the per-target routing of §4. The statement is scoped: it holds for standard CES at 10 ms, from the core to the pedestal top, with the NBI torque unobserved. Edge E_r and edge rotation are known to move within an ELM cycle at sub-millisecond resolution [83, 84], which a 10 ms single-sightline exposure does not resolve, so nothing here speaks to the edge inside an ELM cycle.

6.8 What does the complexity buy, and does size help? A ladder and a width sweep

The model-size and model-family axes are closed; each was measured under the same protocol with its decision rule fixed before TEST scoring. The interpretable rung `b3k8` (§4.6; 21,498 parameters) has a paired T_i skill against the backbone of −0.009, −0.005, +0.026, −0.004 in the cut population (mean +0.002, every CI containing zero) and of −0.16 to −0.21 in the inclusive one (mean −0.194, CI below zero on 4/4 splits), where a bounded correction cannot recover a spiked carried value: rows whose persistence error exceeds 2 keV are 0.6–1.3% of the inclusive test rows and carry 73–83% of `b3k8`'s T_i squared error (70–83% of every other arm's, PCHIP included). Widening the `seq_v2` T_i encoder from 24 to 260 units (34k / 49k / 114k / 358k / 879k parameters, a 26× range) gives mean T_i skill +0.230 / +0.236 / +0.235 / +0.236 / +0.230 (cut), within ±0.008 of the 358k backbone on average, with $V_{\rm rot}$ unchanged (+0.250 to +0.254). Averaging four initializations changes T_i skill by a paired +0.020 in the cut population (CI above zero on 4/4 splits) and +0.007 in the inclusive one (2/4), on the pre-registered 0.02 band and not above it. In an offline model-family card (gap-filling, scored against a bidirectional version of the backbone), a shot-encoding neural process and a transformer pretrained on another tokamak (and the same transformer from a random initialization) change T_i skill by −0.094 to +0.026, no arm clearing the 0.02 band in both populations, and with no same-shot $V_{\rm rot}$ label every arm's skill against the training mean is −0.003 to −0.040.

6.9 The edge concentrates in high-variability neighbourhoods

Using a target-independent, input-side proxy for local activity (large neighbour-bracket slope and/or high local CES-neighbour variance, computed *excluding* the target row), we split each TEST split into "peak" (high-local-activity; 4.1–4.6k T_i rows and 2.4–2.9k $V_{\rm rot}$ rows per split) and bulk rows and score `seq_v2` in each with the shot-clustered bootstrap (Fig. 5):

- T_i: peak-stratum skill$_{\rm PCHIP}$ = +0.45 to +0.61 (cut) and +0.62 to +0.72 (inclusive), **PASS on 8/8**, against an interpolator that has the future neighbour; the bulk carries +0.09 to +0.20 (cut, PASS 4/4) and +0.06 to +0.19 (inclusive, PASS 2/4). Interpolation is near-optimal on the smooth bulk; the model's value is in the active stretches, and that statement is unconditional.

- $V_{\rm rot}$: peak-stratum skill +0.54 to +0.79 (point estimates positive on 8/8, PASS 2/4 in each population, against persistence +0.75 to +0.86 PASS 8/8), and ≈ 0 in the bulk (−0.07 to +0.15, PASS 0/8). The asymmetry is therefore *regional*: on the four-split count the global $V_{\rm rot}$ interval vs. interpolation includes zero on 3/4 and 2/4 splits (the replicate's pooled readout later puts the mean skill above zero, §6.2), but in high-activity neighbourhoods—where smooth past+future interpolation is worst—even the history-driven $V_{\rm rot}$ predictor adds value. We report it as a regional strength with a thin significance base, not a reversal of the global null.

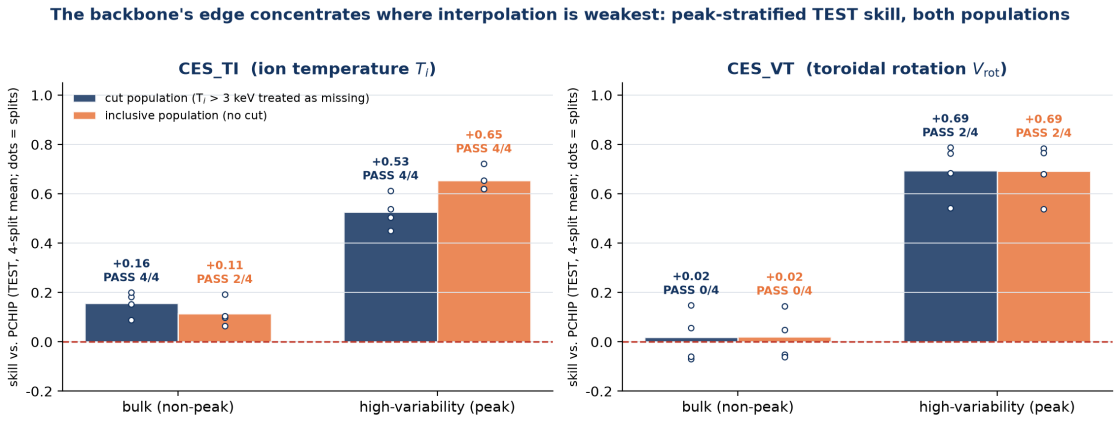


Figure 5: Peak-stratified TEST skill of `seq_v2`, both populations (4-split means, per-split points, PR4 counts). The skill over interpolation concentrates where interpolation is weakest; V_{rot} skill lives entirely in the peak stratum.

Table 9: The error of a typical row. Four test splits pooled (301 shots), cut population, absolute-error quantiles in physical units. M_1 is the median-AE skill against PCHIP (shot-clustered 95% CI); the win rate is the fraction of rows on which the model's error is below PCHIP's. The backbone's T_i median of 47 eV equals the target's own 10 ms bulk scatter (MAD of adjacent-sample differences, 46.4 eV). The Huber arm is the subject of the last paragraph of §6.10.

target	arm	median	q75	q90	M_1 vs PCHIP	win rate
T_i [eV]	`seq_v2` backbone (MSE loss)	47.2	101.0	202.4	−0.198 [−0.259, −0.135]	0.48
	`seq_v2` Huber loss	**43.4**	**93.8**	**193.3**	−0.101	
	persistence	46.1	116.6	262.7		
	causal GP	44.4	106.7	219.8		
	PCHIP (uses the future)	39.4	105.4	236.6		
V_{rot} [km/s]	`seq_v2` backbone (MSE loss)	4.3	10.0	21.4	−0.488 [−0.604, −0.387]	0.40
	`seq_v2` Huber loss	**3.8**	**9.2**	**19.8**	−0.333	
	persistence	3.7	9.3	20.9		
	causal GP	3.8	9.3	20.0		
	PCHIP (uses the future)	2.9	7.8	18.3		

6.10 The aggregate skill is a tail effect: in the median row the nowcaster carries its anchor

An MSE-based skill is decided by a sum of squared errors, and that sum is decided by a few large ones: the backbone's top 1% of T_i squared-error rows carry 48% (cut) and 83% (inclusive) of the total (PCHIP: 51% and 87%). "+0.24 on T_i" is therefore not, by itself, a statement that the model's error is below interpolation's on a *typical* row. To separate the two we re-scored the frozen TEST artifacts, with no retraining, under three robust co-metrics whose rules—and the sentence each outcome would write—were fixed before any number: a median-absolute-error skill $M_1 = 1 - \mathrm{medAE}_{\mathrm{model}}/\mathrm{medAE}_{\mathrm{PCHIP}}$ (shot-clustered weighted-median bootstrap), the skill of a Huber loss clipped at the target's own 10 ms reproducibility bound (T_i 129.9 eV, V_{rot} 16.4 km/s), M_2, and a symmetric 1%-trimmed MSE skill M_3.

The T_i advantage comes from the tail. M_1 against PCHIP is negative, and significantly so, on all 16 population × target × split cells (cut T_i −0.198, inclusive −0.562). On half of the rows the model's error is at or above PCHIP's (win rate 0.48); the quantile ladder of the error ratio model/PCHIP falls from 1.20 at q50 through 1.01 at q70 and 0.96 at q75 to 0.78 at q99—the ratio drops below 1 in the upper error quartile, the fit-failure, transient and ELM rows. In that tail M_2 (+0.156) and M_3 (+0.210) are both significantly positive,

so it is not an effect of the top 1% alone. The median row has a simpler picture: the backbone's 47 eV is persistence's 46 eV and the target's own bulk scatter of 46.4 eV. *On a typical row the nowcaster carries the last observation, noise included*, and PCHIP is 20% better there because averaging with the future neighbour halves that noise—a gain no causal method can have. The bulk ordering (acausal GP 38.5 < PCHIP 39.4 < causal GP 44.4 < persistence 46.1 ≈ backbone 47.2) is exactly the ordering of how many noisy samples each method averages.

The $V_{\rm rot}$ null is not a tail artifact. The pre-registration wrote "ties in aggregate but wins in the median" down as $V_{\rm rot}$'s possible rescue. The opposite was observed: $M_1 = -0.488$ (cut) and -0.518 (inclusive), significantly negative on 4/4 splits, and in the median row the model is worse than persistence (4.3 vs 3.7 km/s). The aggregate interval that includes zero is a bulk deficit offset by a tail gain (peak-stratum +0.54 to +0.79, §6.9), the same conclusion as the term-by-term audit of §6.7.

The loss's influence function moves that bulk. One variable was changed—the influence function of the training loss—with the residual scale δ pinned to the target's own 10 ms reproducibility bound (cut: T_i 129.9 eV, $V_{\rm rot}$ 16.4 km/s; inclusive 274.3 eV): a Huber loss (r^2 inside, $2\delta|r| - \delta^2$ outside), everything else the backbone's. Selected on its own validation objective, the Huber arm's T_i median-AE skill against the MSE backbone is $M_1 = +0.128$ $[+0.114, +0.141]$ in the cut population (CI above zero on 4/4 splits) at a pooled MSE difference of +0.007 with the interval including zero (significant deficits 0/4), with the T_i median at 0.894× persistence; in the inclusive population $M_1 = +0.285$ $[+0.250, +0.322]$ (4/4) but one split keeps a T_i MSE difference of −0.040 with the CI below zero (unchanged when the epoch cap is raised from 60 to 100), so by the pre-registered rule the Huber arm is the **robust rung, conditional on the cut population**, and the backbone keeps the MSE loss. The Huber rows of Table 9 are the same loss selected on validation MSE, which lowers the cut-population median by 8–10% (T_i 47.2 to 43.4 eV, $V_{\rm rot}$ 4.3 to 3.8 km/s) at a paired MSE skill of −0.004 / −0.007 with the interval including zero.

6.11 A physics filter is the median estimator: a dual readout

§6.10 located the headroom in the median row, where the backbone only equals persistence. That row is the regime a filter is built for: a slowly varying level observed through noise, with occasional jumps. We therefore asked whether a *physical* measurement model, not a learned one, owns the median. The cell is a scalar state-space filter per target, run causally on the CES history alone: the prior holds the level and lets it jump at rate λ with jump variance σ_J^2, and the measurement is Student-t ($\nu_0 = 3$) at the target's 10 ms bulk scatter estimated on TRAIN (T_i 0.116 / 0.071 target s.d. in the cut / inclusive population, $V_{\rm rot}$ 0.060). Every parameter is fixed by the registration; training it end-to-end moves none of them by more than 7%, so they are not reported as measured plasma quantities. With Student-t noise at that bulk scale the pure filter's median absolute error is **0.916**× / **0.911**× persistence for T_i and **0.975**× / **0.976**× for $V_{\rm rot}$ (cut / inclusive).

One output cannot carry both objectives. Adding the backbone's MSE-trained residual to the filter ($y = a + r$) meets the physics-rung bar in the cut population (M_1 vs the backbone +0.077 to +0.117, 4/4, MSE deficit ≤ 1/4) and misses the inclusive MSE guard by 0.0014, and in each of the three registered variants—the plain residual, its checkpoint selected on validation median error, and the residual weighted by the filter's variance gain—the combined output gives the filter's median back ($V_{\rm rot}$ 1.07–1.21× persistence), so within this family an output that must also hold the aggregate moves the quiet rows.

The dual readout. The pre-registered consequence is to report the median and the mean from separate estimators on the same rows (Table 10). For T_i the physics filter is the median estimator **unconditionally**: its median skill is +0.06 to +0.39 over the backbone and +0.08 to +0.09 over persistence, CIs above zero on 4/4 splits in both populations, at 0.906–0.920× persistence with a split spread of ±0.007, against ±0.1 or more for every trained arm. For $V_{\rm rot}$ its median skill over the backbone is +0.09 to +0.27 with CIs above zero on 4/4 splits in both populations, and its median-error ratio to persistence is 0.94–1.00× by split with the interval including 1 on 3/4 splits. That interval still includes 1 after fitting the filter on TRAIN.

Table 10: Dual readout on the frozen TEST rows (no retraining). Median estimator: the pure Student-t hold-or-jump filter; mean estimator: the trained model (anchor + residual where its MSE guard holds, i.e. cut; `seq_v2` in the inclusive population). M_1 is the median-AE skill, shot-clustered paired bootstrap; counts are splits with $M_1 \geq +0.05$ and CI > 0 (vs the backbone) or $M_1 > 0$ and CI > 0 (vs persistence). Ranges are over the four splits.

target	population	median / persistence	M_1 vs `seq_v2`	M_1 vs persistence	MSE vs `seq_v2`
T_i	cut	0.911–0.920	+0.06 to +0.14, **4/4**	+0.08 to +0.09, **4/4**	−0.30 to −0.41
	inclusive	0.906–0.915	+0.22 to +0.39, **4/4**	+0.09, **4/4**	−0.35 to −0.47
$V_{\rm rot}$	cut	0.939–1.000	+0.09 to +0.23, **4/4**	0.00 to +0.06, 1/4	−0.06 to −0.51
	inclusive	0.938–1.003	+0.09 to +0.27, **4/4**	0.00 to +0.06, 1/4	−0.06 to −0.52

Likelihood-fitted jump statistics (5–13 Hz) and a likelihood-fitted noise law (Student-t with $\nu \to 2$ and a 2.2 km s^{-1} core) leave the median at 0.97–0.98× persistence, with the interval excluding 1 on at most 2/4 splits. The reason is measured. The 10 ms scatter sits at the photon floor: the ratio of the relative T_i scatter to the Doppler velocity scatter is the $\sqrt{2}$ of a Gaussian line fit in 8/8 split–population cells, and the noise of the two channels is independent. Between labels the filter's predictive is symmetric about the carried value, so its median-optimal point is the last value. The included-1 interval is therefore a limit of the 10 ms CES measurement, not of the estimator. The filter's aggregate MSE skill against the backbone is −0.06 to −0.52 because event rows stay late, which is exactly why it is not the mean estimator. The deployment statement becomes: median from the filter, mean from the nowcaster, and the trade between them measured, not hidden in one output.

6.12 The predictive distribution: one heavy-tailed peak, reported calibrated

A nowcast used for control needs an uncertainty, so the backbone's head was replaced by a predictive distribution trained by its negative log-likelihood (nothing else changes; the posterior mean goes through the same point harness). Three likelihood families were registered in turn: a heteroscedastic Gaussian, a two-component Gaussian mixture, and a Student-t with learnable ν. The reference is the backbone's point prediction with its own TEST RMSE as the scale, an oracle choice that favours the reference. The registration fixed four guards (CRPS gain, central 50 % and 90 % coverage, and no posterior-mean MSE loss against the backbone) and named the arm before any number.

The Gaussian is rejected. Its central 50 % interval covers 0.62–0.80 of the rows (Fig. 6): a single Gaussian scale inflated by the tail rows is too wide for the bulk. On $V_{\rm rot}$ its likelihood weighting is pulled by the $|v| > 1000$ km s^{-1} fit failures and its mean's MSE skill against the backbone is −0.03 to −0.09, with the CI below zero on 3/4 cut splits.

The named arms are calibrated. In the cut population the Student-t head's CRPS difference to the mixture is +0.009, inside the 0.02 tie width, and it is named for having fewer parameters. Its CRPS gain over the reference is +0.229 (T_i) and +0.363 ($V_{\rm rot}$), with the CI above zero on every split, with 50 % / 90 % coverage 0.48–0.51 / 0.91–0.92. In the inclusive population only the mixture passes the guards (+0.444 / +0.351, coverage 0.53–0.55 / 0.90–0.91). The t location there is a redescending estimator: on the > 3 keV fit-failure rows it stays at the bulk and pays their full square in MSE (−0.134 on the peak rows), whereas the mixture moves its mean with the wide component.

What the distribution says about the physics. The mixture never uses two peaks: its two-mode test fires on no row, and the second component is a wide one with weight ≈ 0.07. The fitted t tail is $\nu = 2.8$–3.5 for T_i and sits on the $\nu > 2$ floor for $V_{\rm rot}$. The nowcast error is therefore one heavy-tailed peak: a narrow photon-noise core with rare large departures from events and fit failures, the same shape the physics filter needed for its measurement model (§6.11). The T_i predictive width rises 1.2× on event rows; the $V_{\rm rot}$ width does not grow with the time since the last label, so the $V_{\rm rot}$ uncertainty is a per-discharge scale, not a variance that accumulates between labels. Figure 7 shows the intervals on one held-out discharge.

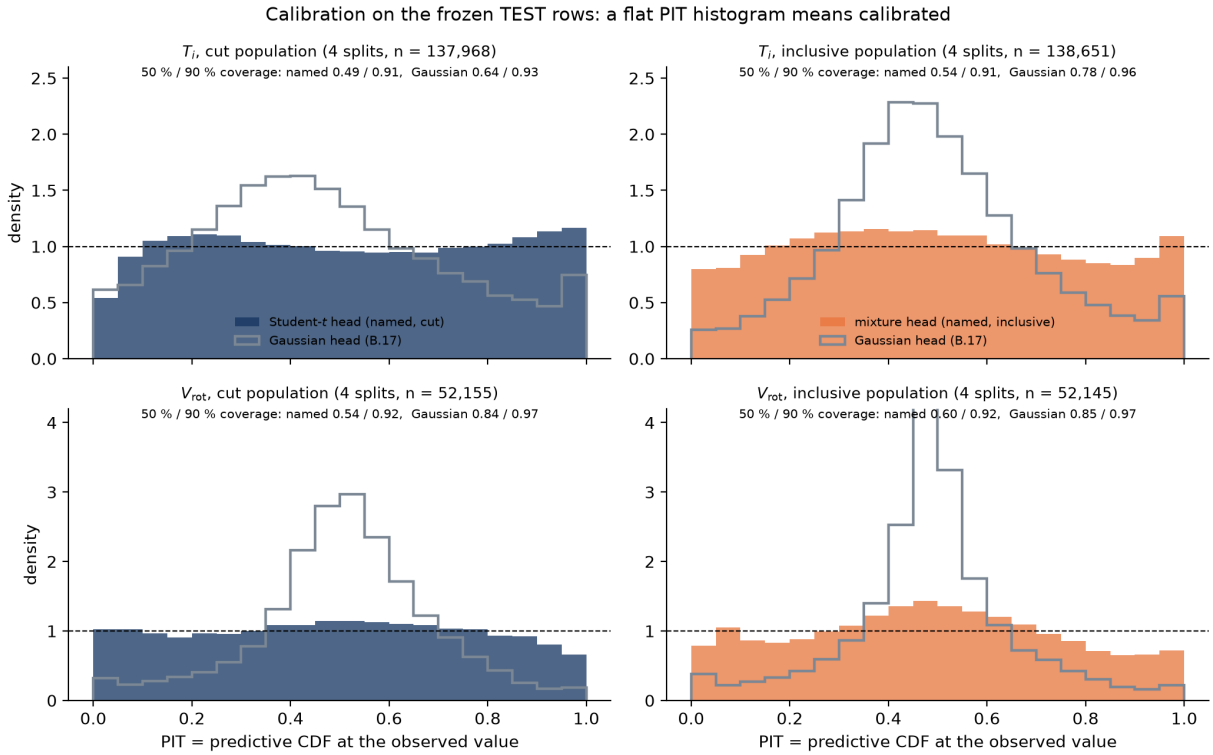


Figure 6: PIT histograms on the frozen TEST rows, four splits pooled. Filled: the registered arm (Student-t in the cut population, mixture in the inclusive one). Outline: the Gaussian head. A flat histogram at density 1 is calibrated; a central hump means intervals that are too wide in the bulk.

6.13 Cut-threshold sensitivity

Retraining the backbone with the spike threshold at 2.5 and 4 keV (four splits each, scored in its own population) gives mean T_i skill +0.230 and +0.232 against +0.236 at 3 keV, PR4 4/4 vs. PCHIP and 4/4 vs. the causal GP at every threshold, and $V_{\rm rot}$ +0.252 / +0.253 / +0.257. The threshold is immaterial across the range a physicist would defend; what matters is the two populations, not where inside the tail the cut falls.

6.14 Context, structure and cost: what sets skill and what sets the price

History length, contiguous context, operator family and cost were moved separately, each under a pre-registered rule. In the windowed family a 24-run sweep ($W \in \{2, 3, 4, 6, 8\}$ and a history-0 point, four split seeds each; hold-free, per-file cap 500, inclusive population) puts the mean T_i skill at -0.026 (PR4 0/4) and $V_{\rm rot}$ at -0.78 with no past CES observation, at +0.238 (4/4) and +0.206 with one, and at +0.190 to +0.246 and +0.190 to +0.206 for every longer window, against a seed spread of 0.07–0.16 within a single point, so the smallest-W-at-plateau rule returns $W = 2$. For the backbone, trained at each context length r on non-overlapping segments (truncating a trained state after the fact reads unlearned state reconstruction as missing information; retraining recovers at least 84% of that deficit), the pre-registered rule—the smallest r whose deficit against full context is under 0.02 with at most one significant deficit out of four splits—returns **5 steps = 50 ms**; pooled over 301 discharges the T_i skill over the causal GP is $+0.057$ $[+0.027, +0.085]$ at 20 ms, $+0.104$ $[+0.079, +0.128]$ at 50 ms and $+0.143$ $[+0.118, +0.168]$ at 630 ms (four-split means +0.055, +0.098 and +0.138), every interval excluding zero, with the per-discharge win rate at 0.52, 0.64 and 0.67. One past observation and 50 ms of contiguous context are different resources, and the windowed family cannot supply the second at any W (its window is pooled without order and its unlabelled rows are dropped): the W=2 control's skill over the causal GP is +0.041 (CI above zero on 1/4 splits) and `seq_v2` restricted to the same two steps has +0.055 (2/4). At matched reach a recurrent LSTM (357,570 parameters), a dilated

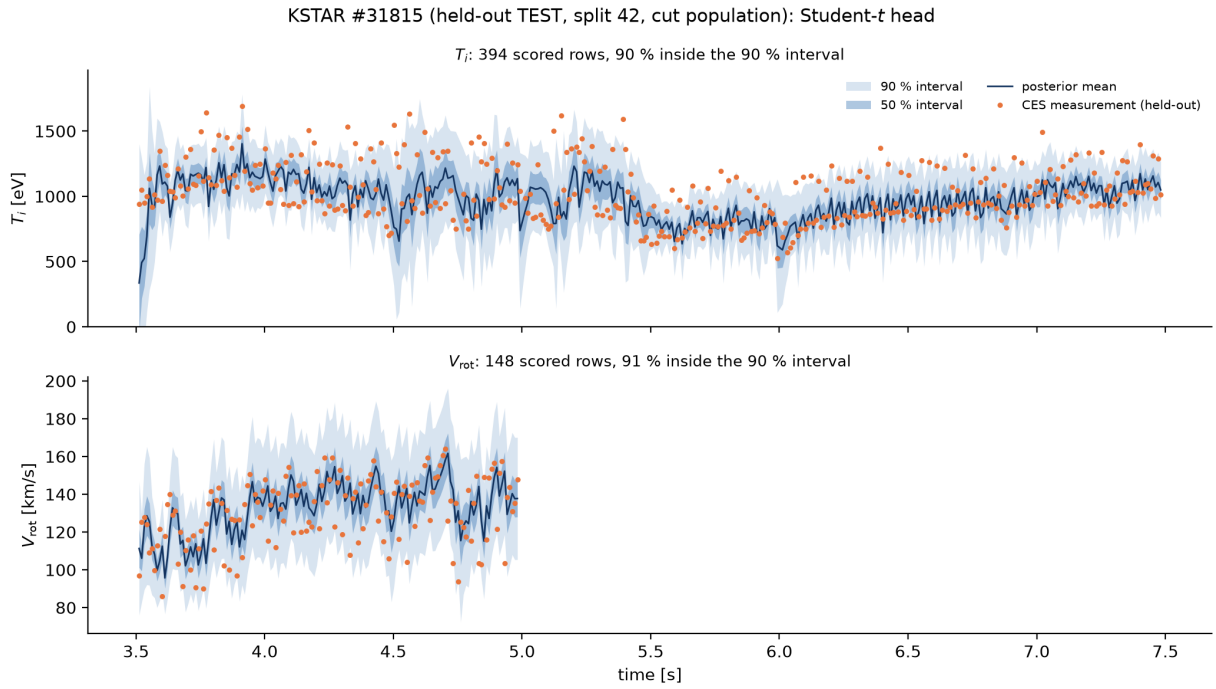


Figure 7: Posterior mean and central 50 % / 90 % predictive intervals of the Student-t head on the held-out discharge #31815 (split 42, cut population), with the CES measurements. The T_i axis is cut at 0; the first row's interval extends below it.

causal convolution (3,238) and banded attention (295,746) have T_i skill over the causal GP of +0.113, +0.119 and +0.119 with the CI above zero on 4/4 splits; the largest paired family difference is −0.023 (attention at the 70 ms rung, CI excluding zero on 3/4 splits), against +0.060 for moving reach from 20 to 70 ms, and a diagonal state-space model reaches +0.105 at 630 ms against the recurrent +0.143 (pooled). Online single-step cost follows the number of dispatched operators (about 2–3 μs each): 111 per step for the LSTM at any reach, 209–305 for the dilated convolution and 473 for banded attention at reach 15–63, so the architecture is chosen on cost and not on skill; measured as the network forward pass only (1,000 timed calls at batch 1 after warm-up, laptop-class machine), the stateful `seq_v2` step costs **0.97 ms median and 1.49 ms at p99 on CPU** (0.51–1.05 ms median in earlier runs of the same benchmark), which is the basis for calling it usable inside a 10 ms cycle.

The model's per-discharge skill is positive where the target moves. On the same context ladder the pooled V_{rot} skill at 70 ms is +0.123 [+0.036, +0.248], excluding zero, but the per-discharge win rate is 0.48 and removing the top five of some 62 discharges takes the pooled skill to zero or below on all four splits (T_i: win rate 0.70, largest single contributor 9–16%), and of eleven covariates tested the one that predicts a positive per-discharge skill is how much the target moves within a discharge: by tercile of that spread the win rate is 42% / 83% / 85% for T_i and 34% / 48% / 55% for V_{rot}.

7 Where the remaining headroom is

The negative results in this paper are unusually specific, and specificity is what makes them useful: they do not say "this problem is hard", they say *which change would move it.* Capacity is ruled out (§6.8), longer windows are ruled out (§6.14), inductive bias, cross-device pretraining and ensembling are ruled out (§6.8), and the framing that buys reach has been adopted (§6.3). The one model-side axis that moved is the loss's influence function, and its gain is a bulk gain (§6.10). What remains beyond that is data, and each lever is identified by evidence already in this dataset.

1. CES fit-quality metadata would replace a proxy with a measurement. The two-population report exists because a value cut is a proxy for "the fit failed": it removes one-sample upward events that no method can predict, but it is one-sided (dips are untouched) and it catches only 19% of $\geq 2\times$ upward outliers (§3.5). With per-sample fit χ^2 or signal level, a quality cut would replace or accompany the value cut in every arm at once, and the two populations would collapse into one. The route to those values already exists on the same device: NN-CES analyses CES spectra without beam modulation and so yields a fit diagnostic per exposure [20], which also removes the gaps that come from modulation itself. This lever is a CES reanalysis product, not a code change on our side. $V_{\rm rot}$'s own spikes (119 rows above 1,000 km/s) would be handled by the same rule; until then any persistence-anchored $V_{\rm rot}$ comparison reports the share of squared error those rows carry, as §6.8 does.

2. The raw Mirnov stream was tested: the registered features do not carry $V_{\rm rot}$. On the identical 10 ms grid, lag-1 autocorrelation within contiguous blocks is +0.568 for BES and +0.572 for ECEI but −0.009 for the Mirnov coils, with 82% of blocks below $|r| = 0.1$: the magnetic channel is white noise *on this grid*. That is the signature of $\mathrm{d}B/\mathrm{d}t$ oscillating at the kHz mode frequency and being decimated to 100 Hz without an anti-aliasing filter, so successive samples have random relative phase. The mode-rotation frequency—the one quantity in our diagnostic set that could proxy toroidal rotation—was therefore discarded upstream of the model, which is also why every derived MC feature we tried (integral, PCHIP integral, $|MC|$, rolling RMS) failed: information already lost cannot be recovered downstream. That reading named a preprocessing fix, and the fix was tested under a pre-registration. The raw 2 MHz coil stream was obtained for 12 discharges, and 15 causal 10 ms window features (RMS envelope and five band powers per coil, and the dominant peak of the two-coil cross-spectrum) were routed into the $V_{\rm rot}$ branch, with six training discharges and three $V_{\rm rot}$-carrying test discharges. The paired change in $V_{\rm rot}$ skill is -0.128 $[-0.544, -0.105]$ in the cut population and -0.164 $[-0.473, -0.133]$ in the inclusive one. Both intervals exclude zero on the negative side, and 0 of 3 test discharges improve. The coil physics itself is confirmed: the edge harmonic oscillation of #32092 appears at 4.39 and 8.91 kHz. But the dominant mode frequency in a 10 ms window is not a row-by-row function of the CES edge rotation: its within-discharge Spearman correlation with $V_{\rm rot}$ is $|\rho| \leq 0.11$ in six of seven discharges. The rejection covers the 15 registered features on a six-discharge training set. Other functions of the raw stream (for example the cross-phase slope over many coils) were not tested. Raw Mirnov features are therefore not a named next experiment for the 10 ms $V_{\rm rot}$ nowcast.

3. The actuator channel is absent, and at 10 ms it would change little. Toroidal rotation is set by injected torque, and no torque signal exists in this dataset. The data show the break precisely: between shots, the ECE-derived T_e surrogate correlates with T_i at $r = +0.353$ ($p = 3 \times 10^{-17}$) but with $V_{\rm rot}$ at $r = +0.024$ ($p = 0.58$), and within shots the T_i correlation keeps a consistent sign while the $V_{\rm rot}$ one is sign-random. So T_e does track the heating level, and heating level does not reach rotation—power is not torque, which depends on beam energy, tangency radius and injection geometry. NBI torque is the cause of the rotation *level*. It is not a lever for the 10 ms nowcast. Beam power is a step input, held constant between a handful of source on/off steps per discharge, so it acts only on the slow band. The part of the 10 ms increment variance carried by a 300 ms moving average is 0.34% for $V_{\rm rot}$ (interquartile range 0.21–0.94%, 20 discharges) and 0.25% for T_i (63 discharges). A perfect torque channel therefore bounds the removable 10 ms increment variance at about 1%, and the level it sets is already in the input as the value observed 10 ms earlier. The channel matters for a different scope: forecasts of 100 ms and longer after a power step, and the cold start before the first CES sample. The positive control in the literature is of that kind: recurrent full-discharge simulators on DIII-D that receive the beam actuators as inputs predict rotation's evolution [85].

4. The rest of the bulk belongs to the future neighbour. After the Huber arm's recovery, a 4 eV median gap to PCHIP remains (§6.10). Its size is what averaging two adjacent noisy observations buys, so the only way a causal model closes it is by denoising the history—weighting it by the per-exposure fit quality a CES re-analysis would provide—which is the same deliverable as lever 1. And the zero-label row of the model-family card (§6.8) shows the cold-start scope of item 3: with no same-shot $V_{\rm rot}$ label, no learned arm rises above the training mean from BES, ECEI and Mirnov alone.

For the 10 ms nowcast the lever that remains is therefore the per-exposure fit quality of items 1 and 4, which the 641 shot files do not contain. Item 2 is a route closed within its registered scope, and item 3 belongs

to forecasts of 100 ms and longer.

8 Limitations

The headline skill is a tail statistic. In the median row the backbone equals persistence and trails PCHIP by 20%, and V_{rot} is below persistence in the median (§6.10); the paper therefore reports median absolute error beside every aggregate skill and does not claim a lower error than interpolation on a typical row. The scale-pinned Huber loss recovers 8–10% of the bulk at no aggregate cost (13% when selected on its own objective), reported as conditional on the cut population; the same rule lifts the inclusive median gain to +0.285 but leaves one split with an MSE difference of −0.040 whose CI is below zero, which a longer cap does not remove (§6.10). The T_i median is owned by a physics filter unconditionally (§6.11), reported beside, not inside, the aggregate model; for V_{rot} that filter's median-error ratio to persistence is 0.94–1.00 with the interval including 1 on 3/4 splits, at the photon floor of the 10 ms line fit. **Statistical power.** The unit of replication is the shot; with 96 (T_i) / 60–66 (V_{rot}) test shots per split, power bounds every significance call, and per-shot squared-error differences are heavy-tailed (a few discharges dominate the bootstrap spread; in the inclusive population ≈1% of rows carry 70–83% of every arm's T_i squared error). **MNAR optimism.** Skill is measured on observed points only; the reweighting of §6.5 bounds the optimism for the causal comparison in both populations and for the offline comparison with population-conditional significance, and it covers 54–68% of missing T_i rows and 4–6% of missing V_{rot} rows. **The offline claim is bounded above by the offline GP.** The offline GP's T_i RMSE is below PCHIP's (153.8 vs. 173.6) and the model's paired skill against it is −0.05 to +0.11 with the CI including zero on 7/8 cells, so "skill over future-using interpolation" means "skill over the pre-registered interpolants"; it does not extend to every offline method. It does extend to the causal GP, the strongest deployable one (+0.08 to +0.17, CIs above zero on 8/8). **The value cut is a proxy.** Two populations bracket the fit-failure question; a fit-quality cut would settle it (§7), and V_{rot}'s spikes remain uncut. **Campaign transfer rests on one temporal block** (§6.6): four initialisations, not four splits, and 2/4 cut-population runs stopped at the epoch cap. The repair that makes the windowed control transfer, per-shot standardization, is the offline form; a causal running (expanding-window/EWMA) estimator is the version that would ship and is not measured here. **Metric asymmetry.** Interpolants use full-shot neighbours while the windowed control sees two rows and the backbone the segment's past; this is deliberately adverse to the causal models but complicates direct interpretation. **Single device and model family.** Everything is one device, one diagnostic set and two closely related recurrent families; nothing here establishes transfer to another tokamak. **Uncertainty is calibrated on pooled rows.** The coverage of the predictive head is reported per split over all rows (§6.12); its coverage per discharge was not measured. **Latency** is a network-only measurement on one laptop-class machine, with up to 2× run-to-run variation from power states (§6.14); feature assembly and acquisition overhead are not included. **Scope.** The pedestal-top framing is inherited from the data selection (pedestal-top CES channels); no radial-dependence experiment is performed, and event-phase analyses (ELM cycle, L–H transitions) are future work—the high-variability analysis of §6.9 is an event-agnostic proxy. **Scope of the asymmetry.** The $T_i \leftrightarrow V_{\mathrm{rot}}$ statement holds for standard CES at 10 ms, from the core to the pedestal top, with the NBI torque unobserved; edge E_r and rotation are known to move within an ELM cycle [83, 84], and a measurement that resolves the ELM cycle could carry fast-channel information about rotation that ours cannot. **The equation is three quarters of the backbone, read in one population.** 22–28% of the backbone's PCHIP-relative skill is nonlinear event handling that the 20-parameter form does not reproduce, and its physical bands are read from the cut population only: in the inclusive population the equation keeps 0.43–0.81 of the PCHIP-relative skill and its BES gain changes sign under the fit-failure spikes (§4.4).

9 Conclusion

A missing 10 ms CES row in KSTAR can be filled with lower error, for ion temperature, by a strictly causal estimator than by interpolation that reads the future. We built that estimator—a full-grid two-branch causal LSTM whose V_{rot} branch never sees the fast diagnostics—and evaluated it under a pre-registered, shot-clustered, two-population protocol against a deliberately adverse bar: interpolation that sees the future, and a causal GP that is the strongest method that does not. The model's skill over future-using PCHIP for

ion temperature is +0.17 to +0.32 across four independent splits and both populations with every interval excluding zero ([+0.19, +0.25] and [+0.19, +0.31] under a pre-registered repeated 5-fold replicate that tests every discharge once), its paired skill against the strongest offline smoother is −0.05 to +0.11 with the interval including zero on 7/8 cells, and its skill over the causal GP is +0.08 to +0.17 with the interval above zero on all eight cells (and +0.27 and +0.21 in the non-adjacent regime, $\Delta t > 15$ ms). The skill survives both stress tests: at genuinely missing, in-domain timesteps the skill over persistence stays at +0.28 to +0.44 with intervals above zero on 8/8 cells, and across a campaign boundary the sequence nowcaster keeps +0.17 to +0.20 over PCHIP and +0.11 to +0.16 over the causal GP with intervals above zero on 4/4 initialisations, where the windowed control it replaced had lost its offline advantage (interval above zero on 2/4 and 0/4 vs. PCHIP)—the fast diagnostics drift 5–11× more between campaigns than the targets, and the sequence framing standardizes per shot and reaches the whole past by construction.

What that nowcaster computes can be written down. Weight magnitude told us nothing stable; the input–output response did, and a four-stage registered readout turned it into a 20-parameter physical filter: an observation filter whose gain on a new measurement rises with the gap since the last one, plus a linear response to each fast channel's deviation from its preceding 100 ms baseline that is switched off beyond a learned threshold. Trained from scratch under the identical protocol, that equation keeps 72–78% of the backbone's PCHIP-relative skill on 4/4 cut splits, above the registered 0.70 bar. The remainder is the network's nonlinear event handling; it is the price of a 20-parameter description, not a missing term that these inputs expose. The bulk of the skill is denoising of the noisy anchor, and the margin over interpolation is carried by the rows where a pedestal event is seen in BES and ECEI before the next CES exposure.

The same equation explains the converse result. For rotation the fast channels carry nothing the model uses—zeroing them leaves the V_{rot} output bit-identical—so its mean-squared-error skill over interpolation is a tail effect carried by history alone (95% interval [+0.07, +0.70] cut vs. PCHIP under the registered two-level bootstrap, excluding zero), and in the median the interval of its error ratio to persistence includes 1 on 3/4 splits. The physics is the one our inputs do not observe: the NBI torque that drives rotation, and the photon floor of a 10 ms line fit. The statement is scoped to standard CES at 10 ms from the core to the pedestal top; edge rotation inside an ELM cycle is a different measurement.

Two estimands are kept apart. The aggregate skill is a tail statistic—in the median row the nowcaster equals persistence—and a pre-registered hold-or-jump filter with Student-t noise at the training bulk scale is the median estimator (0.91× persistence for T_i on 8/8 cells); every output that must also hold the aggregate gives that median back, so median and mean are reported from separate estimators, and a Student-t distribution head carries the calibrated uncertainty (§6.12).

We also report what is not claimed and what would overturn it. The offline claim is bounded above by the offline GP; the value cut that defines one population is a one-sided proxy for fit quality, and CES fit-quality metadata would replace it; the reweighting reaches 54–68% of missing T_i rows and a twentieth of missing V_{rot} rows; the predictive intervals are calibrated on pooled rows, and per-discharge coverage was not measured; beyond 45 ms the inclusive-population interval includes zero (−0.004, [−0.30, +0.25]); the equation's physical bands are read from the cut population only, since under the fit-failure spikes its inclusive BES gain changes sign; and everything is one device and one campaign. The model-size and model-family axes are closed—a 21k-parameter persistence-plus-bounded-correction model matches the backbone under the cut, a 26× width sweep is flat, and a shot-encoding neural process, a transformer pretrained on another tokamak and a 4-init ensemble change T_i skill by −0.09 to +0.03 with none clearing the 0.02 band in both populations—so what remains is on the input side: per-exposure fit uncertainty from the CES analysis for the 10 ms nowcast, and the NBI torque channel for forecasts of 100 ms and longer. The registered features of the raw 2 MHz Mirnov stream were tested and changed V_{rot} skill by −0.128 [−0.544, −0.105] (§7). The evaluation that supports all of this—pre-registration with a scored prediction ledger, discharge-cluster intervals, overlap-corrected repeated cross-validation and a two-level bootstrap—is assembled from standard pieces of other fields; we found no precedent for the combination in peer-reviewed fusion machine learning (§2), and it is reported as the condition of belief, not as a result.

Reproducibility

Splits are pinned to disk and verified on reload; normalization statistics are train-file-only (plus per-shot input standardization for the sequence family); model selection never reads test data; bootstrap uses a fixed seed

and 10,000 resamples; every quoted number is read from frozen evaluation artifacts by one collection script.

Code availability

The full pipeline, the pre-registration document with its per-batch execution records, the controlled-experiment runners behind every result reported here, the pinned train/validation/test splits, and the collector that regenerates every quoted number from the frozen evaluation artifacts are openly available at https://github.com/iseungsang01/ml-intern-revision. The backbone is `ces_prediction/experiments/seq/model_seq_v2.py`, the interpretable rung `model_seq_b3.py`, and the windowed control `ces_prediction/model_iter009.py` (guarded by a content hash in the test suite), so the code that produced these numbers cannot drift from the code that is released.

Data availability

The KSTAR diagnostic data underlying this study — 641 discharges, shots 30801–32751 — are not redistributed with the code. They are experimental data of the KSTAR device and are governed by the operating institution's data policy; access requests should be directed there. The repository contains everything needed to reproduce the analysis once the shot files are in place: the exact file-level split manifests, the environment configuration for every reported run, and the per-sample squared-error archives from which the bootstrap statistics are computed.

References

[1] W.H. Ko, Hyungho Lee, D. C. Seo, and Myeun Kwon. Charge exchange spectroscopy system calibration for ion temperature measurement in KSTAR. *Review of Scientific Instruments*, 81:10D740, 2010. doi: 10.1063/1.3496991.

[2] Roderick J. A. Little and Donald B. Rubin. *Statistical Analysis with Missing Data*. Wiley Series in Probability and Statistics. Wiley, 2nd edition, 2002. doi: 10.1002/9781119013563.

[3] F. N. Fritsch and R. E. Carlson. Monotone piecewise cubic interpolation. *SIAM Journal on Numerical Analysis*, 17(2):238–246, 1980. doi: 10.1137/0717021.

[4] C. E. Rasmussen and C. K. I. Williams. *Gaussian Processes for Machine Learning*. MIT Press, 2006. ISBN 0-262-18253-X.

[5] M. A. Chilenski, M. Greenwald, Y. Marzouk, N. T. Howard, A. E. White, J. E. Rice, and J. R. Walk. Improved profile fitting and quantification of uncertainty in experimental measurements of impurity transport coefficients using Gaussian process regression. *Nuclear Fusion*, 55(2):023012, 2015. doi: 10.1088/0029-5515/55/2/023012.

[6] C. Michoski, T. A. Oliver, D. R. Hatch, A. Diallo, M. Kotschenreuther, D. Eldon, M. Waller, R. Groebner, and A. O. Nelson. A Gaussian process guide for signal regression in magnetic fusion. *Nuclear Fusion*, 64 (3):035001, 2024. doi: 10.1088/1741-4326/ad1af5.

[7] Minseok Kim, W.H. Ko, Sehyun Kwak, Semin Joung, Wonjun Lee, B. Kim, D. Kim, J.H. Lee, Choongki Sung, Yong-Su Na, and Y.-C. Ghim. Kinetic profile inference with outlier detection using support vector machine regression and Gaussian process regression. *Nuclear Fusion*, 64:106052, 2024. doi: 10.1088/1741-4326/ad7304.

[8] R. L. Dougherty, A. S. Edelman, and J. M. Hyman. Nonnegativity-, monotonicity-, or convexity-preserving cubic and quintic Hermite interpolation. *Mathematics of Computation*, 52(186):471–494, 1989. doi: 10.1090/S0025-5718-1989-0962209-1.

[9] C. H. Reinsch. Smoothing by spline functions. *Numerische Mathematik*, 10(3):177–183, 1967. doi: 10.1007/BF02162161.

[10] G. Kvernadze. Convergence and Gibbs' phenomenon in cubic spline interpolation of discontinuous functions. *Journal of Computational and Applied Mathematics*, 92(2):173–188, 1998. doi: 10.1016/S0377-0427(97)00199-4.

[11] S. Eyheramendy, F. Elorrieta, and W. Palma. An irregular discrete time series model to identify residuals with autocorrelation in astronomical light curves. *Monthly Notices of the Royal Astronomical Society*, 481(4):4311–4322, 2018. doi: 10.1093/mnras/sty2487.

[12] F. Elorrieta, S. Eyheramendy, and W. Palma. Discrete-time autoregressive model for unequally spaced time-series observations. *Astronomy & Astrophysics*, 627:A120, 2019. doi: 10.1051/0004-6361/201935560.

[13] W. H. Press, S. A. Teukolsky, W. T. Vetterling, and B. P. Flannery. *Numerical Recipes: The Art of Scientific Computing*. Cambridge University Press, 3rd edition, 2007.

[14] A. H. Murphy. Skill scores based on the mean square error and their relationships to the correlation coefficient. *Monthly Weather Review*, 116(12):2417–2424, 1988. doi: 10.1175/1520-0493(1988)116<2417:SSBOTM>2.0.CO;2.

[15] Joseph Abbate, Emiliano Fable, Giovanni Tardini, Rainer Fischer, and Egemen Kolemen. Combining physics-based and data-driven models for quantitatively accurate plasma profile prediction. *Nuclear Fusion*, 65:056014, 2025. doi: 10.1088/1741-4326/adc283.

[16] M. J. Owens, R. Challen, J. Methven, E. Henley, and D. R. Jackson. A 27 day persistence model of near-Earth solar wind conditions: A long lead-time forecast and a benchmark for dynamical models. *Space Weather*, 11(5):225–236, 2013. doi: 10.1002/swe.20040.

[17] Chris Bishop, C.M. Roach, and M. G. von Hellermann. Automatic analysis of JET charge exchange spectra using neural networks. *Plasma Physics and Controlled Fusion*, 35:765–773, 1993. doi: 10.1088/0741-3335/35/6/010.

[18] J. Svensson, M. von Hellermann, and R. König. Analysis of JET charge exchange spectra using neural networks. *Plasma Physics and Controlled Fusion*, 41:315–338, 1999. doi: 10.1088/0741-3335/41/2/016.

[19] Baoyue Chai, Yingying Li, Ze Chen, Wei Tao, Yixuan Zhou, Shifeng Mao, Zhengping Luo, Yi Yu, Bo Lyu, and Minyou Ye. Fast estimation of ion temperature from EAST charge exchange recombination spectroscopy using neural network. *Plasma Science and Technology*, 21:105103, 2019. doi: 10.1088/2058-6272/ab2674.

[20] J.K. Lee, W.H. Ko, H.H. Lee, B. Kim, Y.H. Lee, G.W. Shin, J. Kim, and J.M. Lee. Neural network-based analysis of charge exchange spectra in KSTAR. *Fusion Engineering and Design*, 222:115518, 2026. doi: 10.1016/j.fusengdes.2025.115518.

[21] A. Pavone, J. Svensson, A. Langenberg, U. Höfel, S. Kwak, N. Pablant, and R. C. Wolf. Neural network approximation of Bayesian models for the inference of ion and electron temperature profiles at W7-X. *Plasma Physics and Controlled Fusion*, 61:075012, 2019. doi: 10.1088/1361-6587/ab1d26.

[22] Mark D. Boyer and Jason D. Chadwick. Prediction of electron density and pressure profile shapes on NSTX-U using neural networks. *Nuclear Fusion*, 61:046024, 2021. doi: 10.1088/1741-4326/abe08b.

[23] Zichao Lin, Hongming Zhang, Fudi Wang, Cheonho Bae, Jia Fu, Yongcai Shen, Shuyu Dai, Yifei Jin, Dian Lu, Shengyu Fu, Huajian Ji, and Bo Lyu. Prediction of plasma rotation velocity and ion temperature profiles in EAST Tokamak using artificial neural network models. *Nuclear Fusion*, 64:106061, 2024. doi: 10.1088/1741-4326/ad73e8.

[24] Francisco A. Matos, Diogo R. Ferreira, and P. Carvalho. Deep learning for plasma tomography using the bolometer system at JET. *Fusion Engineering and Design*, 114:18–25, 2016. doi: 10.1016/j.fusengdes.2016.11.006.

[25] Cong Wang, Weizhe Yang, Haiping Wang, Renjie Yang, Jing Li, Zhijun Wang, Yixiong Wei, Xianli Huang, Chenshu Hu, Zhaoyang Liu, Xinyao Yu, Changqing Zou, and Zhifeng Zhao. Physics-informed deep learning model for line-integral diagnostics across fusion devices. *Nuclear Fusion*, 65:076022, 2025. doi: 10.1088/1741-4326/ade0ce.

[26] Xishuo Wei, Shuying Sun, W. M. Tang, Zhihong Lin, Hongfei Du, and Ge Dong. Reconstruction of tokamak plasma safety factor profile using deep learning. *Nuclear Fusion*, 63:086020, 2023. doi: 10.1088/1741-4326/acdf00.

[27] G.H. Zheng, Zongyu Yang, Songfen Liu, Rui Ma, Xinwen Gong, Ao Wang, Shuo Wang, and W.L. Zhong. Real-time equilibrium reconstruction by multi-task learning neural network based on HL-3 tokamak. *Nuclear Fusion*, 64:126041, 2024. doi: 10.1088/1741-4326/ad8014.

[28] S. Joung, J. Kim, S. Kwak, J. Bak, S. Lee, H. Han, H. Kim, G. Lee, D. Kwon, and Y. Ghim. Deep neural network Grad-Shafranov solver constrained with measured magnetic signals. *Nuclear Fusion*, 2020.

[29] Semin Joung, Young chul Ghim, Jaewook Kim, S. Kwak, Daeho Kwon, Chih-Jen Sung, Dong Ju Kim, Hyun-Seok Kim, J.G. Bak, and S.W. Yoon. GS-DeepNet: mastering tokamak plasma equilibria with deep neural networks and the Grad–Shafranov equation. *Scientific Reports*, 13:15799, 2023. doi: 10.1038/s41598-023-42991-5.

[30] L. Jung, Jaewook Kim, and Jisung Kang. Machine learning-based plasma profile reconstruction in KSTAR. *Journal of the Korean Physical Society*, 88:1079 – 1087, 2026. doi: 10.1007/s40042-026-01580-1.

[31] Joseph Abbate, Rory Conlin, and Egemen Kolemen. Data-driven profile prediction for DIII-D. *Nuclear Fusion*, 61:046027, 2021. doi: 10.1088/1741-4326/abe08d.

[32] Ricardo Shousha, Jaemin Seo, Keith Erickson, Z. Xing, S.K. Kim, Joseph Abbate, and Egemen Kolemen. Machine learning-based real-time kinetic profile reconstruction in DIII-D. *Nuclear Fusion*, 64:026006, 2024. doi: 10.1088/1741-4326/ad142f.

[33] Semin Joung, Jaewook Kim, S. Kwak, Kyeo reh Park, S.H. Hahn, Hyunsun Han, Hyun-Seok Kim, J.G. Bak, S. G. Lee, and Young chul Ghim. Imputation of faulty magnetic sensors with coupled Bayesian and Gaussian processes to reconstruct the magnetic equilibrium in real time. *Review of Scientific Instruments*, 89:10K106, 2018. doi: 10.1063/1.5038938.

[34] Debashish Mohapatra, Bidyadhar Subudhi, and Raju Daniel. Real-time sensor fault detection in Tokamak using different machine learning algorithms. *Fusion Engineering and Design*, 151:111401, 2019. doi: 10.1016/j.fusengdes.2019.111401.

[35] S. Kaldas, R. Rossi, I. Wyss, M. Gelfusa, and P. Gaudio. Developing autoencoder models for multi-diagnostic electron density processing in tokamaks within a scientific machine learning framework. *Journal of Fusion Energy*, 45, 2026. doi: 10.1007/s10894-026-00568-w.

[36] M.R. Wang, Chenguang Wan, Jingjing Lu, Zhi Yu, Bingjia Xiao, Yanlong Li, X.X. He, Zhengping Luo, Qiping Yuan, Yemin Hu, and Jiangang Li. Time series extrinsic regression for reconstructing missing electron temperature in tokamak. *Nuclear Fusion*, 65:076008, 2025. doi: 10.1088/1741-4326/addb5f.

[37] K. Jia, Z. Xu, Z. Wu, W. Gao, M. Wang, A. Ti, et al. Time series regression network using exogenous variables for missing effective ion charge reconstruction in EAST tokamak. *Nuclear Fusion*, 67:016012, 2026. doi: 10.1088/1741-4326/aeb432.

[38] Zongyu Yang, Zhenghao Yang, Wenjing Tian, Jiyuan Li, Xiang Sun, Guohui Zheng, Songfen Liu, Niannian Wu, Rongpeng Li, Zhaohe Xu, Bo Li, Zhongbing Shi, Zhe Gao, Wei Chen, Xiaoquan Ji, Min Xu, and Wulyu Zhong. FusionMAE, a self-supervised pretrained model to optimize and simplify diagnostic and control of fusion plasma. *Communications Physics*, 2026. doi: 10.1038/s42005-026-02626-3.

[39] Azarakhsh Jalalvand, S.K. Kim, Jaemin Seo, Qiming Hu, M. Curie, Peter Steiner, A. Nelson, Yong-Su Na, and Egemen Kolemen. Multimodal super-resolution: discovering hidden physics and its application to fusion plasmas. *Nature Communications*, 16:8506, 2025. doi: 10.1038/s41467-025-63492-1.

[40] M. Imríšek, J. Seidl, Michal Odložilík, M. Šos, P. Bílková, Petr Boehm, Haluk Sait Aksoy, Raul Scapini Weiand, Iván Alcaraz Aguado, Robin Mahieu, and D. Naydenkova. Temporal super-resolution of kinetic plasma profiles at COMPASS using neural networks. *Plasma Physics and Controlled Fusion*, 68:065049, 2026. doi: 10.1088/1361-6587/ae72c6.

[41] F. Pastore, F. Felici, T. O. S. J. Bosman, C. Galperti, O. Sauter, B. Vincent, and N. M. T. Vu. Model-based electron density estimation using multiple diagnostics on TCV. *Fusion Engineering and Design*, 192:113615, 2023. doi: 10.1016/j.fusengdes.2023.113615.

[42] HyeongJun Noh, Chweeho Heo, Xiaotian Gao, and Yong-Su Na. PanoMHD: Multimodal modelling of plasma dynamics towards Tokamak control. 2026.

[43] Zhengping Che, Sanjay Purushotham, Kyunghyun Cho, David Sontag, and Yan Liu. Recurrent neural networks for multivariate time series with missing values. *Scientific Reports*, 8:6085, 2018. doi: 10.1038/s41598-018-24271-9.

[44] Satya Narayan Shukla and Benjamin M. Marlin. Multi-time attention networks for irregularly sampled time series. 2021:14897–14911, 2021. doi: 10.48550/arxiv.2101.10318.

[45] Yulia Rubanova, Ricky T. Q. Chen, and David Duvenaud. Latent ordinary differential equations for irregularly-sampled time series. 32:5320–5330, 2019.

[46] Rich Caruana. Multitask learning. *Machine Learning*, 28:41–75, 1997. doi: 10.1023/a:1007379606734.

[47] Jiquan Ngiam, Aditya Khosla, Mingyu Kim, Juhan Nam, Honglak Lee, and Andrew Y. Ng. Multimodal deep learning. In *Proceedings of the 28th International Conference on Machine Learning (ICML)*, 2011.

[48] Tadas Baltrušaitis, Chaitanya Ahuja, and Louis-Philippe Morency. Multimodal machine learning: A survey and taxonomy. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 41:423–443, 2018. doi: 10.1109/tpami.2018.2798607.

[49] Ruibin Xiong, Yunchang Yang, Di He, Kai Zheng, Shuxin Zheng, Xing Chen, Huishuai Zhang, Yanyan Lan, Liwei Wang, and Tie-Yan Liu. On layer normalization in the transformer architecture. In *Proceedings of the 37th International Conference on Machine Learning (ICML)*, 2020.

[50] Maximilian Ilse, Jakub M. Tomczak, and Max Welling. Attention-based deep multiple instance learning. 2018. doi: 10.48550/arxiv.1802.04712.

[51] Xingjian Shi, Zhourong Chen, Hao Wang, Dit-Yan Yeung, Wai Kin Wong, and Wang-chun Woo. Convolutional LSTM network: A machine learning approach for precipitation nowcasting. 2015. doi: 10.48550/arxiv.1506.04214.

[52] Suman Ravuri, Karel Lenc, Matthew Willson, Dmitry Kangin, Rémi Lam, Piotr Mirowski, M. A. Fitzsimons, Maria Athanassiadou, Sheleem Kashem, Sam Madge, Rachel Prudden, Amol Mandhane, Aidan Clark, Andrew Brock, Karen Simonyan, Raia Hadsell, N. Robinson, Ellen Clancy, Alberto Arribas, and Shakir Mohamed. Skilful precipitation nowcasting using deep generative models of radar. *Nature*, 597:672–677, 2021. doi: 10.1038/s41586-021-03854-z.

[53] Brian A. Nosek and Daniël Lakens. Registered reports: A method to increase the credibility of published results. *Social Psychology*, 45(3):137–141, 2014. doi: 10.1027/1864-9335/a000192.

[54] Luca Bertinetto, João F. Henriques, Samuel Albanie, Michela Paganini, and Gül Varol, editors. *Proceedings of the NeurIPS 2020 Workshop on Pre-registration in Machine Learning*, volume 148 of *Proceedings of Machine Learning Research*, 2021. PMLR. URL https://proceedings.mlr.press/v148/.

[55] Claude Nadeau and Yoshua Bengio. Inference for the generalization error. *Machine Learning*, 52:239–281, 2003. doi: 10.1023/A:1024068626366.

[56] Stephen Bates, Trevor Hastie, and Robert Tibshirani. Cross-validation: What does it estimate and how well does it do it? *Journal of the American Statistical Association*, 119(546):1434–1445, 2024. doi: 10.1080/01621459.2023.2197686.

[57] Varun Saravanan, Gordon J. Berman, and Samuel J. Sober. Application of the hierarchical bootstrap to multi-level data in neuroscience. *Neurons, Behavior, Data analysis, and Theory*, 3(5):1–25, 2020. doi: 10.1101/819334. Preprint: bioRxiv 819334.

[58] Shiquan Ren, Hong Lai, Wenjing Tong, Mostafa Aminzadeh, Xuezhang Hou, and Shenghan Lai. Nonparametric bootstrapping for hierarchical data. *Journal of Applied Statistics*, 37(9):1487–1498, 2010. doi: 10.1080/02664760903046102.

[59] Thibault Sellam, Steve Yadlowsky, Jason Wei, Naomi Saphra, Alexander D'Amour, Tal Linzen, Jasmijn Bastings, Iulia Turc, Jacob Eisenstein, Dipanjan Das, Ian Tenney, and Ellie Pavlick. The MultiBERTs: BERT reproductions for robustness analysis. In *International Conference on Learning Representations (ICLR)*, 2022. arXiv:2106.16163.

[60] L. Goddard, A. Kumar, A. Solomon, D. Smith, G. Boer, P. Gonzalez, V. Kharin, W. Merryfield, C. Deser, S. J. Mason, B. P. Kirtman, R. Msadek, R. Sutton, E. Hawkins, T. Fricker, G. Hegerl, C. A. T. Ferro, D. B. Stephenson, G. A. Meehl, T. Stockdale, R. Burgman, A. M. Greene, Y. Kushnir, M. Newman, J. Carton, I. Fukumori, and T. Delworth. A verification framework for interannual-to-decadal predictions experiments. *Climate Dynamics*, 40(1–2):245–272, 2013. doi: 10.1007/s00382-012-1481-2.

[61] Tilmann Gneiting and Adrian E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007. doi: 10.1198/016214506000001437.

[62] Manuel Gebetsberger, Jakob W. Messner, Georg J. Mayr, and Achim Zeileis. Estimation methods for nonhomogeneous regression models: Minimum continuous ranked probability score versus maximum likelihood. *Monthly Weather Review*, 146(12):4323–4338, 2018. doi: 10.1175/MWR-D-17-0364.1.

[63] Stephan Rasp and Sebastian Lerch. Neural networks for postprocessing ensemble weather forecasts. *Monthly Weather Review*, 146(11):3885–3900, 2018. doi: 10.1175/MWR-D-18-0187.1.

[64] David Sussillo and Omri Barak. Opening the black box: Low-dimensional dynamics in high-dimensional recurrent neural networks. *Neural Computation*, 25(3):626–649, 2013. doi: 10.1162/neco_a_00409.

[65] Miles Cranmer, Alvaro Sanchez-Gonzalez, Peter Battaglia, Rui Xu, Kyle Cranmer, David Spergel, and Shirley Ho. Discovering symbolic models from deep learning with inductive biases. In *Advances in Neural Information Processing Systems 33 (NeurIPS 2020)*, 2020. arXiv:2006.11287.

[66] Niru Maheswaranathan and David Sussillo. How recurrent networks implement contextual processing in sentiment analysis. arXiv:2004.08013, 2020.

[67] Mark Rothmann, Ning Li, Gang Chen, George Y. H. Chi, Robert Temple, and Hsiao-Hui Tsou. Design and analysis of non-inferiority mortality trials in oncology. *Statistics in Medicine*, 22(2):239–264, 2003. doi: 10.1002/sim.1400.

[68] Yann LeCun, John S. Denker, and Sara A. Solla. Optimal brain damage. In *Advances in Neural Information Processing Systems 2 (NIPS 1989)*, pages 598–605. Morgan Kaufmann, 1990.

[69] L. Allen and C. M. Bishop. Neural network approach to energy confinement scaling in tokamaks. *Plasma Physics and Controlled Fusion*, 34(7):1291–1302, 1992. doi: 10.1088/0741-3335/34/7/008.

[70] Michael Roth, Emre Özkan, and Fredrik Gustafsson. A Student's t filter for heavy tailed process and measurement noise. In *2013 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 5770–5774, 2013. doi: 10.1109/ICASSP.2013.6638770.

[71] Gabriel Agamennoni, Juan I. Nieto, and Eduardo M. Nebot. Approximate inference in state-space models with heavy-tailed noise. *IEEE Transactions on Signal Processing*, 60(10):5024–5037, 2012. doi: 10.1109/TSP.2012.2208106.

[72] Katharina Rath, David Rügamer, Bernd Bischl, Udo von Toussaint, Cristina Rea, Andrew Maris, Robert Granetz, and Christopher G. Albert. Data augmentation for disruption prediction via robust surrogate models. *Journal of Plasma Physics*, 88(5):895880502, 2022. doi: 10.1017/S0022377822000769.

[73] H. van den Brand, M. R. de Baar, M. van Berkel, T. C. Blanken, F. Felici, E. Westerhof, M. Willensdorfer, The ASDEX Upgrade Team, and The EUROfusion MST1 Team. A model-based, multichannel, real-time capable sawtooth crash detector. *Plasma Physics and Controlled Fusion*, 58(7):075002, 2016. doi: 10.1088/0741-3335/58/7/075002.

[74] Yuya Morishita, Sadayoshi Murakami, Masayuki Yokoyama, and Genta Ueno. Data assimilation system based on integrated transport simulation of Large Helical Device plasma. *Nuclear Fusion*, 60(5):056001, 2020. doi: 10.1088/1741-4326/ab7596.

[75] Zander Keith, Chirag Nagpal, Cristina Rea, and R. Alex Tinguely. Risk-aware framework development for disruption prediction: Alcator C-Mod and DIII-D survival analysis. *Journal of Fusion Energy*, 43(1): 21, 2024. doi: 10.1007/s10894-024-00413-y.

[76] Y. U. Nam, S. Zoletnik, M. Lampert, and Á. Kovácsik. Analysis of edge density fluctuation measured by trial KSTAR beam emission spectroscopy system. *Review of Scientific Instruments*, 83:10D531, 2012. doi: 10.1063/1.4739078.

[77] G.S. Yun, W. Lee, M. Choi, Joon Bum Kim, HK Park, C. W. Domier, Benjamin Tobias, T. Liang, Xiangyu Kong, N. C. Luhmann, and A. J. H. Donné. Development of KSTAR ECE imaging system for measurement of temperature fluctuations and edge density fluctuations. *Review of Scientific Instruments*, 81:10D930, 2010. doi: 10.1063/1.3483209.

[78] Committee for Proprietary Medicinal Products (CPMP). Points to consider on switching between superiority and non-inferiority. European Medicines Agency, CPMP/EWP/482/99, 2000.

[79] International Council for Harmonisation (ICH). Addendum on estimands and sensitivity analysis in clinical trials to the guideline on statistical principles for clinical trials E9(R1). ICH Harmonised Guideline E9(R1), Step 4, 2019.

[80] Brandon D. Gallas, Andriy Bandos, Frank W. Samuelson, and Robert F. Wagner. A framework for random-effects ROC analysis: Biases with the bootstrap and other variance estimators. *Communications in Statistics - Theory and Methods*, 38(15):2586–2603, 2009. doi: 10.1080/03610920802610084.

[81] Art B. Owen. The pigeonhole bootstrap. *The Annals of Applied Statistics*, 1(2):386–411, 2007. doi: 10.1214/07-AOAS122.

[82] J. E. Rice, A. Ince-Cushman, J.S. deGrassie, L.-G. Eriksson, Y. Sakamoto, A. Scarabosio, A. Bortolon, K.H. Burrell, B.P. Duval, C. Fenzi-Bonizec, M. Greenwald, R. J. Groebner, G. T. Hoang, Y. Koide, E. S. Marmar, A. Pochelon, and Y. Podpaly. Inter-machine comparison of intrinsic toroidal rotation in tokamaks. *Nuclear Fusion*, 47:1618–1624, 2007. doi: 10.1088/0029-5515/47/11/025.

[83] E. Viezzer, M. Cavedon, P. Cano-Megias, E. Fable, E. Wolfrum, D. J. Cruz-Zabala, et al. Dynamics of the pedestal transport during edge localized mode cycles at ASDEX Upgrade. *Plasma Physics and Controlled Fusion*, 62:024009, 2020. doi: 10.1088/1361-6587/ab5b1d.

[84] M. Cavedon, T. Pütterich, E. Viezzer, F. M. Laggner, A. Burckhart, M. G. Dunne, R. Fischer, A. Lebschy, et al. Pedestal and E_r profile evolution during an edge localized mode cycle at ASDEX Upgrade. *Plasma Physics and Controlled Fusion*, 59:105007, 2017. doi: 10.1088/1361-6587/aa7ad0.

[85] Ian Char, Youngseog Chung, Joseph Abbate, Egemen Kolemen, and Jeff Schneider. Full shot predictions for the DIII-D tokamak via deep recurrent networks. arXiv:2404.12416, 2024.